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# Path-Gain Compressibility Determines Local Credit Assignment in Conductance-Based Dendrites

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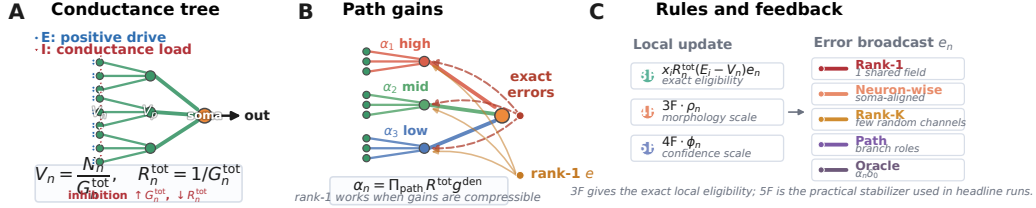
## Abstract

1 Biologically plausible credit assignment requires replacing the compartment-  
2 specific errors of backpropagation with signals available to dendritic synapses.  
3 We study when a low-dimensional somatic broadcast can serve as such a substitute  
4 in conductance-based dendritic trees. We derive exact gradients showing that each  
5 synaptic update factorizes into a local eligibility term, presynaptic activity times  
6 driving force times input resistance, and a single non-local compartment error.  
7 This exposes the key approximation made by local learning: replacing the exact  
8 path-specific compartment error with a broadcast field. We show that shunting  
9 inhibition changes this approximation by modulating input resistance along den-  
10 dritic paths and, when it preferentially attenuates high-gain paths, compresses the  
11 effective path-gain field. Across matched additive and shunting dendritic networks,  
12 shunting improves local-backprop gradient alignment and reduces gradient scale  
13 mismatch. Transported-error oracle experiments show that supplying the exact  
14 path-gain-transported error largely closes the local-learning gap, identifying broad-  
15 cast fidelity as the dominant bottleneck. Under nonnegative conductances and  
16 rank-1 broadcast, shunting LocalCA approaches matched backpropagation ceilings  
17 on MNIST, Fashion-MNIST, and context-gating, while routed and harder-data  
18 diagnostics reveal when richer pathway-specific feedback is required. These results  
19 identify path-gain compressibility as a mechanistic condition under which simple  
20 local credit assignment is effective in dendritic networks.

## 21 1 Introduction

22 Credit assignment in deep networks usually depends on backpropagation: global error transport  
23 through symmetric pathways with no obvious biological substrate. Dendritic compartments suggest  
24 an alternative. Each synapse has access to rich local state such as driving force, membrane conduc-  
25 tance, and branch voltage, while supervisory information could arrive as a low-bandwidth broadcast  
26 from the soma [14]. The question here is not whether such a broadcast can match unconstrained  
27 backpropagation in all settings, but when it is a faithful approximation to the exact dendritic credit  
28 signal.

29 Starting from conductance-based dendritic voltage equations [1], we derive exact gradients for  
30 dendritic trees (Theorem 1) and show that each synaptic gradient factorizes into purely local eligibility  
31 terms and a single non-local compartment error. Approximating that compartment error with a  
32 broadcast signal yields a theorem-facing local rule (3F), practical optimizer wrappers (4F/5F), and  
33 higher-rank feedback variants for tasks where different branches require different teaching signals.



**Figure 1: Model and credit assignment.** (A) Conductance-based dendritic tree. Nonnegative excitatory and inhibitory inputs drive distal compartments, which feed proximal branches and then the soma; shunting inhibition enters the conductance denominator and lowers input resistance. (B) Path gains through the tree. Exact backprop supplies path-specific compartment errors, while rank-1 LocalCA replaces that field with one shared broadcast. (C) Local rule and feedback choices. The 3F rule contains the exact local eligibility from the theorem; 4F/5F add stabilizing local factors, and the feedback ladder ranges from rank-1 broadcast to transported oracle.

34 The central finding is mechanistic: shunting can make low-bandwidth broadcast useful by improving  
 35 the compressibility of the conductance-stage path-gain field. Dendritic branching creates multiple  
 36 error paths from distal synapses to the soma. Excitatory and inhibitory synapse banks set the conduc-  
 37 tance state of each compartment. Shunting makes inhibitory conductance enter the denominator, so  
 38 inhibition changes input resistance and therefore the gain of every upstream path that passes through  
 39 that compartment. This does not mean inhibition always helps. It helps rank-1 broadcast when it  
 40 attenuates high-gain paths more than low-gain paths, narrowing or low-rank compressing the exact  
 41 error field that the broadcast must approximate.

42 We quantify this chain with three diagnostics. First, exact-gradient reconstruction verifies that  
 43 the mathematical factorization matches autograd to numerical precision. Second, gradient-fidelity  
 44 measurements compare local and backprop gradients at matched parameters. Third, transported-error  
 45 oracle experiments replace the broadcast with the exact path-gain-transported compartment error,  
 46 isolating broadcast fidelity from other architectural factors. Across matched networks, shunting  
 47 improves local-backprop alignment on MNIST (cosine 0.202 vs. 0.006) and greatly reduces scale  
 48 mismatch (0.117 vs. 1.053). On noise resilience, transported error raises additive local learning from  
 49 66.8% under rank-1 broadcast to 92.4–94.0%, nearly matching shunting at 94.3–95.2%.

50 **Contributions.** We make three claims: (i) exact conductance-tree credit factorizes into synapse-  
 51 local eligibility and a path-specific compartment error (Theorem 1, Corollary 1); (ii) path-gain  
 52 compressibility is a measurable mediator of when rank-1 broadcast approximates that compartment  
 53 error, and shunting provides a biophysical route to such compression by changing input resistance  
 54 along dendritic paths; and (iii) gradient-fidelity, transported-oracle, competence, and morphology  
 55 experiments validate this mechanism while identifying the failure modes that require richer feedback.

56 The empirical story follows this sequence. Figure 1 defines the conductance tree, exact credit object,  
 57 and broadcast taxonomy. Figure 2 verifies the exact factorization and measures local-backprop fidelity.  
 58 Figure 3 links path-gain compressibility to broadcast fidelity, oracle transport, and learning. Figures 4  
 59 and 5 show the practical supervised regimes and the morphology-dependent inhibitory operating  
 60 point. Harder-data and routed-feedback experiments are reported as appendix diagnostics rather than  
 61 headline benchmark claims.

## 2 Compartmental Voltage Model and Gradient Derivation

63 We use a steady-state conductance model derived from discretized passive cable dynamics [1, 2]. In  
 64 normalized units, leak and inhibitory reversal are set to 0, the excitatory reversal is set to 1, and leak  
 65 conductance is fixed at 1. Each compartment voltage is therefore a conductance-weighted average.  
 66 Two properties of this form matter for what follows: local sensitivities depend on both driving force  
 67 ( $E - V$ ) and input resistance  $R^{\text{tot}}$ , and shunting inhibition corresponds to adding conductance when  
 68  $E_{\text{inh}} \approx 0$ . Throughout the paper we write dendritic morphologies as rooted trees  $[b_1, b_2, \dots, b_D]$ ,  
 69 where each factor gives the fan-in from one dendritic stage to the next on the path to the soma. Thus  
 70  $[3, 3]$  means that each soma aggregates 3 proximal branches and each proximal branch aggregates 3  
 71 distal branches, for 9 distal leaves per soma. In code, each excitatory unit is a stack of branch layers,

each branch layer carries explicit excitatory and optional inhibitory synapse banks plus a branch-to-parent aggregation map, and the headline feedforward regime keeps `somatic_synapses=false` so inputs terminate on dendritic branches rather than directly on the soma. The main experiments use `input_mode=1`: the transfer layer supplies nonnegative excitatory and inhibitory input streams, and inhibition enters through learned branch-level I-to-E conductance banks. This is input-driven shunting inhibition, not a recurrent lateral inhibitory circuit. The additive and shunting cores are therefore architecture-matched at the tree level; they differ only in the branch-level voltage integration rule. These ingredients play distinct roles in the credit-assignment argument. Branching creates the path structure. E/I synapse banks determine the local conductance state. Shunting turns inhibitory conductance into a multiplicative change in input resistance. The local learning rule then asks whether a low-dimensional teaching signal can stand in for the exact path-specific error induced by that tree.

## 2.1 Voltage Equation and Local Sensitivities

Consider compartment  $n$  with synaptic inputs  $j$  (activity  $x_j$ , reversal  $E_j$ , conductance  $g_j^{\text{syn}} \geq 0$ ) and dendritic inputs from children (voltage  $V_j$ , conductance  $g_j^{\text{den}} \geq 0$ ). The steady-state voltage is:

$$V_n = \frac{\sum_j E_j x_j g_j^{\text{syn}} + \sum_j V_j g_j^{\text{den}}}{\underbrace{\sum_j x_j g_j^{\text{syn}} + \sum_j g_j^{\text{den}} + 1}_{g_n^{\text{tot}}}}, \quad R_n^{\text{tot}} = 1/g_n^{\text{tot}}. \quad (1)$$

$V_n$  is a convex combination of reversal potentials, child voltages, and leak. Writing  $\mathcal{S}_n = \{0\} \cup \{E_j\}_j \cup \{V_j\}_j$ , we have  $\min \mathcal{S}_n \leq V_n \leq \max \mathcal{S}_n$  and  $0 < R_n^{\text{tot}} \leq 1$ . The local sensitivities follow directly:

**Proposition 1** (Local Sensitivities).

$$\frac{\partial V_n}{\partial g_i^{\text{syn}}} = x_i R_n^{\text{tot}} (E_i - V_n), \quad \frac{\partial V_n}{\partial V_i} = g_i^{\text{den}} R_n^{\text{tot}}, \quad \frac{\partial V_n}{\partial g_i^{\text{den}}} = R_n^{\text{tot}} (V_i - V_n). \quad (2)$$

## 2.2 Shunting Inhibition as Divisive Gain Control

An inhibitory synapse with  $E_{\text{inh}} \approx 0$  contributes current  $(0 - V_n)x_j g_j^{\text{syn}}$  and increases  $g_n^{\text{tot}}$ . Its sensitivity is  $\partial V_n / \partial g_j^{\text{syn}} = -x_j R_n^{\text{tot}} V_n$ , which corresponds to multiplicative attenuation (divisive normalization). Shunting is divisive at the voltage level, but its effect on firing rates can be subtractive in some regimes [7]. Inhibitory plasticity can balance excitation dynamically [8]; our learned inhibitory conductances serve an analogous role. The same denominator also gives the direct connection to path gain. Writing  $G_n^E(x)$  and  $G_n^I(x)$  for the total excitatory and inhibitory synaptic conductance impinging on compartment  $n$ ,

$$R_n^{\text{tot}}(x) = (1 + G_n^E(x) + G_n^I(x) + G_n^{\text{den}})^{-1}.$$

Increasing  $G_n^I(x)$  lowers  $R_n^{\text{tot}}(x)$  and therefore multiplicatively suppresses every upstream path whose error must pass through compartment  $n$ . In this sense inhibition can gate credit flow by changing the path gain, even when the inhibitory drive is feedforward rather than lateral. Prop. 2 formalizes this statement after the tree path gain is defined.

## 2.3 Exact Gradients for Dendritic Trees

Let  $V_0$  be the somatic output and define the soma/core error as  $\delta_0 := \partial L / \partial V_0$ . For a linear decoder  $\hat{y} = W_{\text{dec}} V_0$ , this reduces to  $\delta_0 = W_{\text{dec}}^\top (\partial L / \partial \hat{y})$ ; for a nonlinear decoder,  $\delta_0$  is the corresponding decoder-input Jacobian product.

**Theorem 1** (Conductance-Stage Backpropagation on a Dendritic Tree). *For a rooted dendritic tree with soma at node 0 and unique parent  $p(n)$  for each non-somatic compartment, if each branch passes its pre-reactivation voltage directly to its parent, then the loss gradient satisfies*

$$\frac{\partial L}{\partial V_n} = \frac{\partial L}{\partial V_{p(n)}} R_{p(n)}^{\text{tot}} g_{n \rightarrow p(n)}^{\text{den}}, \quad (3)$$

with boundary condition  $\partial L / \partial V_0 = \delta_0$ . Because the graph is a tree, each compartment has a unique path to the soma. Defining the path gain

$$\alpha_n = \prod_{(i \rightarrow k) \in \text{path}(n \rightsquigarrow 0)} R_k^{\text{tot}} g_{i \rightarrow k}^{\text{den}}, \quad \frac{\partial L}{\partial V_n} = \alpha_n \delta_0, \quad (4)$$

with  $\alpha_0 = 1$ .

*Proof.* Apply the chain rule on the tree-structured computation graph using Prop. 1.  $\square$

**Proposition 2** (Inhibitory control of conductance-stage path gain). *Let  $\mathcal{A}(n)$  be the set of compartments on the path from compartment  $n$  to the soma, excluding  $n$  and including the parent compartments through which the error must pass. Holding dendritic coupling conductances fixed, an added inhibitory conductance  $\Delta G_k^I(x) \geq 0$  at any  $k \in \mathcal{A}(n)$  changes the conductance-stage path gain by*

$$\frac{\alpha_n(x; \Delta G^I)}{\alpha_n(x; 0)} = \prod_{k \in \mathcal{A}(n)} \frac{g_k^{\text{tot}}(x)}{g_k^{\text{tot}}(x) + \Delta G_k^I(x)}, \quad \frac{\partial \log \alpha_n}{\partial G_k^I} = \begin{cases} -R_k^{\text{tot}}, & k \in \mathcal{A}(n), \\ 0, & k \notin \mathcal{A}(n). \end{cases} \quad (5)$$

*Proof.* Substitute  $R_k^{\text{tot}} = 1/g_k^{\text{tot}}$  into the path-gain product in Eq. 4. Adding inhibitory conductance changes only the denominator at compartments on the path from  $n$  to the soma, which gives the product ratio and the log derivative.  $\square$

Prop. 2 makes two directed predictions. First, inhibition is a path gate: it attenuates all credit paths below the inhibited compartment. Second, inhibition is useful for rank-1 local learning only when this attenuation makes the distribution of  $\alpha_n$  more uniform or better aligned with the task. If inhibition is too strong, too heterogeneous, or generated by a poorly calibrated inhibitory population, it can suppress useful signals and hurt learning. In the additive control the same I input changes voltages and local eligibilities, but it does not enter the conductance-stage path recursion as a multiplicative resistance gate.

**Proposition 3** (When shunting compresses path gains). *Let  $z_n = \log \alpha_n$  be the log path gain for compartment  $n$ . If added inhibitory conductance contributes a path-dependent attenuation*

$$\beta_n(x) = \sum_{k \in \mathcal{A}(n)} \log \left( 1 + \frac{\Delta G_k^I(x)}{g_k^{\text{tot}}(x)} \right), \quad z'_n = z_n - \beta_n, \quad (6)$$

then, over compartments or examples,

$$\text{Var}(z') = \text{Var}(z) + \text{Var}(\beta) - 2 \text{Cov}(z, \beta). \quad (7)$$

Thus shunting narrows the log path-gain field exactly when

$$\text{Cov}(z, \beta) > \frac{1}{2} \text{Var}(\beta). \quad (8)$$

*Proof.* Eq. 6 follows by taking the logarithm of the product ratio in Prop. 2. Eq. 7 is the variance identity for  $z - \beta$ , and rearranging gives the condition.  $\square$

Prop. 3 is the mechanism we test empirically. Shunting is not assumed to concentrate gains by default; it concentrates or low-rank compresses the credit field only when inhibitory attenuation is aligned with the high-gain paths. This is why the morphology and input-mode diagnostics are reported as operating-regime tests rather than as monotonic “more inhibition is better” claims.

**Corollary 1** (Local–Global Factorization). *The exact synaptic gradient at compartment  $n$  factorizes as:*

$$\frac{\partial L}{\partial g_i^{\text{syn}}} = \underbrace{x_i R_n^{\text{tot}} (E_i - V_n)}_{\text{synapse-local eligibility}} \cdot \underbrace{\frac{\partial L}{\partial V_n}}_{\text{compartment error}}, \quad (9)$$

where the eligibility term depends only on quantities available at synapse  $i$  (presynaptic activity  $x_i$ , input resistance  $R_n^{\text{tot}}$ , driving force  $E_i - V_n$ ), and the compartment error  $\partial L / \partial V_n$  is the sole non-local quantity.

Under identity upward transfer, this factorization isolates the only non-local quantity as  $\partial L/\partial V_n = \alpha_n \delta_0$ . With a post-voltage activation  $a_n = f_n(V_n)$  transmitted to the parent, the exact recursion becomes  $\partial L/\partial V_n = f'_n(V_n) (\partial L/\partial V_{p(n)}) R_{p(n)}^{\text{tot}} g_{n \rightarrow p(n)}^{\text{den}}$ , so the compartment error remains the sole non-local term but the effective path gain acquires local activation-derivative factors. Any practical local rule therefore depends entirely on how well a broadcast signal approximates that exact compartment error, which we evaluate in Sec. 4.2. In the models studied here, shunting improves this approximation by reducing and stabilizing  $R_n^{\text{tot}}$ , which narrows the spread of local sensitivities across the tree. The theorem is stated for the pre-activation compartment voltage  $V_n$  produced by the conductance stage itself. In the empirical networks, each dendritic branch layer may optionally apply a post-voltage reactivation  $f(V_n)$  before passing activity to the next stage; unless otherwise noted, the manuscript sweeps use a learnable bounded `param_tanh` reactivation. Our exact-error diagnostics are computed on the hooked pre-activation voltage, so the factorization check and compartment-error analyses target the conductance-stage quantity defined above rather than the post-activation activity.

### 3 Local Learning Rules

#### 3.1 Broadcast Error Approximation

We approximate the exact compartment error  $\partial L/\partial V_n$  (Corollary 1) with a broadcast signal  $e_n$  derived from the somatic/core error  $\delta_0$ . For a linear readout this reduces to  $\delta_0 = W_{\text{dec}}^\top (\partial L/\partial \hat{y})$ ; for a nonlinear decoder we use the corresponding decoder-input Jacobian product  $J_{\text{dec}}(h_{\text{core}})^\top (\partial L/\partial \hat{y})$  so that the broadcast remains defined in the same core-space coordinates. We distinguish five feedback objects:

- **Rank-1 shared broadcast:**  $e_n = \bar{\delta} \cdot \mathbf{1}$ , where  $\bar{\delta} = \text{mean}(\delta_0)$  is one scalar field delivered to all compartments.
- **Neuron-wise broadcast:**  $e_n = \delta_0$  when the layer width matches the core/output error dimension. In our headline feedforward settings this condition usually fails, so the default historical “per-soma” implementation reduces to rank-1 shared broadcast.
- **Rank- $K$  random broadcast:**  $c = P_K \delta_0 \in \mathbb{R}^K$  and  $e_n = \sum_{k=1}^K q_{n,k} c_k$ , where  $P_K$  is a fixed random projection and  $q_{n,k}$  mixes those  $K$  channels into compartment  $n$ .
- **Path-structured broadcast:**  $e_n = \alpha \bar{\delta} \cdot \mathbf{1} + (1-\alpha) \Gamma_n(\delta_0)$ , where  $\Gamma_n$  projects the somatic error into router-inferred pathway roles, gates those roles by local pathway activity, and transports the resulting vector hierarchically through the block-structured dendritic tree.
- **Transported oracle:**  $e_n^{\text{transport}} = \alpha_n \delta_0 = \partial L/\partial V_n$ . This is an analysis upper bound, not a proposed local rule, because it supplies the exact path gain  $\alpha_n$  from Theorem 1.

We use the historical `per_soma` implementation for most experiments, but in the main feedforward architectures it is effectively rank-1 because hidden widths exceed the decoder error dimension (Appendix Table S18). Under shunting, that deliberately low-bandwidth field is often sufficient on standard classification tasks. Routed tasks and rank-bridge controls in the appendix test the opposite regime, where rank-1 broadcast collapses branch identity and richer feedback is needed. A local-mismatch heuristic is included only as an appendix control because it performs much worse than the somatic-error broadcasts (Appendix A).

#### 3.2 Three-Factor Rule (3F)

**Definition 1** (3F Update). *For synaptic and dendritic conductances:*

$$\Delta g_j^{\text{syn}} = \eta \langle x_j R_n^{\text{tot}} (E_j - V_n) e_n \rangle_B, \quad \Delta g_j^{\text{den}} = \eta \langle R_n^{\text{tot}} (V_j - V_n) e_n \rangle_B, \quad (10)$$

where  $\langle \cdot \rangle_B$  denotes the batch average.

The three factors are presynaptic activity  $x_j$  (or a voltage difference), postsynaptic modulation through driving force and input resistance, and the broadcast error  $e_n$ . The same rule applies to excitatory and inhibitory synapses. The sign difference comes from the driving force  $(E_j - V_n)$ .

188 **Additive control.** Rule (10) is the local gradient of the *shunting* voltage  $V_n = \sum E_j g_j x_j / g_n^{\text{tot}}$ . For  
 189 the additive control ( $V_n = \sum g_j x_j$ , no divisive normalization), the correct local gradient is simpler:  
 190  $\Delta g_j^{\text{syn}} = \eta \langle x_j e_n \rangle_B$ , with no driving-force or  $R_n^{\text{tot}}$  terms ( $R_n^{\text{tot}} \equiv 1$  by definition since there is no  
 191 denominator). Throughout, each architecture uses the learning rule derived from its own forward-pass  
 192 dynamics. This keeps the comparison architecture-matched rather than applying a shunting-derived  
 193 rule to an additive model.

### 194 3.3 Practical Heuristic Wrappers: 4F and 5F

195 **4F (heuristic attenuation proxy).** In the conductance-stage recursion (4), the path gain  $\alpha_n$  attenuates  
 196 the somatic error differently at each compartment. To compensate without explicitly computing that  
 197 gain, 4F uses an online correlation proxy

$$\rho_n = \text{Cov}(\bar{V}_n, \bar{V}_0) / (\sqrt{\text{Var}(\bar{V}_n) \text{Var}(\bar{V}_0)} + \varepsilon).$$

198 High  $\rho_n$  up-weights branches whose voltages covary with the soma; low  $\rho_n$  down-weights branches  
 199 where the broadcast is likely to be a poor proxy.

200 **5F (confidence-weighted 4F).** 5F adds a second heuristic factor

$$\phi_n = \text{Var}(V_n) / (\sigma_{\text{res}}^2 + \varepsilon),$$

201 where  $\sigma_{\text{res}}^2$  is the residual variance of  $V_n$  after linear regression on the parent voltage. Operationally,  
 202  $\phi_n$  is a branch-wise confidence gate: it becomes large when local voltages are strong and stable  
 203 relative to unexplained residual fluctuations, and small when local fluctuations dominate. We clamp  
 204 it to  $[0.25, 4.0]$  for stability. This factor is empirical, not theorem-derived. The paper’s core claim  
 205 rests on the exact factorization, path-gain compressibility, and the transported-oracle experiment; the  
 206 gain from  $\phi_n$  should be read as practical denoising rather than theoretical necessity.

**Proposition 4 (5F Update).**

$$\Delta g_j^{\text{syn}} = \eta \rho_n \phi_n \langle x_j R_n^{\text{tot}} (E_j - V_n) e_n \rangle_B. \quad (11)$$

207 A feedback-alignment-style random-broadcast argument is included in Appendix C; it is supportive  
 208 but not central to the main mechanism claim.

### 209 3.4 Structured pathway feedback and branch-role plasticity

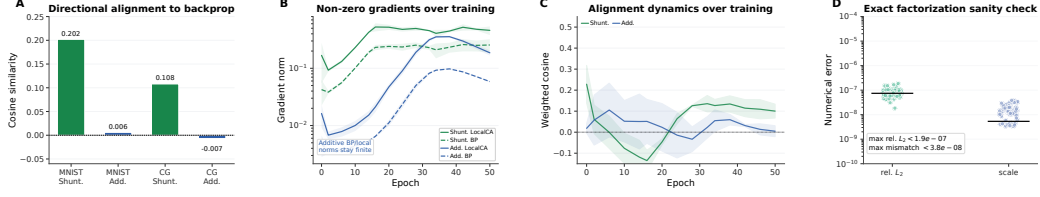
210 For tasks with latent pathway structure, scalar broadcast can erase the distinction between different  
 211 dendritic branches that should be credited differently. Recent experimental evidence supports the  
 212 existence of vectorized instructive signals in cortical dendrites [43], motivating the structured feedback  
 213 variant we introduce here. We therefore augment the local rule with a structured cue-routing variant,  
 214 *PV-LocalCA*.

215 **Hierarchical pathway-vector broadcast.** The encoder/router defines a low-dimensional role basis  
 216 over latent pathways. Instead of collapsing  $\delta_0$  to a rank-1 or neuron-wise broadcast, we project it into  
 217 this role basis and gate each role by local pathway activity:

$$e_n^{\text{PV}} = \sum_{k=1}^K q_{n,k} c_k,$$

218 where  $c_k = \langle r_k, \delta_0 \rangle$  are role coefficients and  $q_{n,k}$  are local branch-role weights. Scalar broadcast is  
 219 the special case  $K=1$ , while PV-LocalCA is a structured rank- $K$  extension rather than a separate  
 220 learning principle.

221 **Router-derived branch roles.** Rather than assigning hand-coded apical/basal labels, each branch  
 222  $j$  receives a normalized role profile  $r_j$  inferred from router assignments and incoming block weights.  
 223 Plasticity is then gated by branch selectivity and by a synapse-specific role-alignment factor  $\langle r_i, r_j \rangle$ ,  
 224 biasing learning toward pathway-consistent afferents without imposing a fixed branch taxonomy.



**Figure 2: Gradient-fidelity diagnostic.** (A) Weighted cosine similarity between local and backprop gradients: shunting (green) is consistently more aligned than additive (blue) on both MNIST and context gating. (B) Full-vector local and backprop gradient norms over training. Additive gradients remain finite, so the near-zero additive cosine in panel A reflects a directional mismatch rather than vanishing gradients. (C) 5F alignment dynamics over training. Shunting remains more positively aligned, whereas additive stays near zero. (D) Exact-factorization sanity check across all theory-diagnostic conditions, reported with relative  $L_2$  reconstruction error and scale mismatch. Reconstructed gradients match autograd to numerical precision, showing that LocalCA’s only approximation is in the broadcast substitute for the compartment error.

## 225 4 Experiments

### 226 4.1 Setup

227 We evaluate two settings: a *capacity-calibrated* regime, where matched architectures achieve strong  
 228 backprop performance, and *controlled sweeps* that isolate inhibition strength, broadcast mode,  
 229 morphology, and architecture. The main text uses MNIST, Fashion-MNIST [37], context gating,  
 230 and noise resilience; cue integration and CIFAR-10 are reported as appendix diagnostics because  
 231 they are most useful for probing failure modes rather than establishing the core mechanism. Formal  
 232 task definitions are in Appendix I. Architectures include point MLP baselines and dendritic cores  
 233 with either additive integration or shunting. Main comparisons are architecture-matched additive vs.  
 234 shunting cores and local vs. backprop training; DFA appears only as a supplementary baseline. Most  
 235 MNIST, Fashion-MNIST, and context-gating results use 5F with the historical per\_soma broadcast  
 236 implementation. Because the hidden widths in these feedforward settings exceed the decoder output  
 237 dimension, that implementation reduces to rank-1 shared broadcast in the main mechanism and  
 238 competence sweeps; higher-rank controls then use explicit low-rank, path-propagation, path-transport,  
 239 or pathway-vector variants. Unless otherwise stated, dendritic layers also use the learnable post-  
 240 voltage reactivation param\_tanh; Appendix Tables S11 and S12 then audit activation shape in a  
 241 more controlled setting by comparing identity (none) against a frozen param\_tanh under matched  
 242 additive/shunting initialization. All dendritic runs supporting the main claims keep synaptic and  
 243 dendritic conductances nonnegative via positive parameter transforms and enforce a nonnegative  
 244 first-layer synaptic drive. Image inputs are used in [0, 1], noise-resilience corruption is clipped back  
 245 to [0, 1], and context gating uses a nonnegative contextual figure-ground MNIST construction. The  
 246 inhibition sweeps additionally use a transported-error oracle derived from Theorem 1 as an upper  
 247 bound rather than as the proposed biological rule. Main mechanistic, oracle, competence, morphology,  
 248 and activation-audit results use 5 seeds per condition unless otherwise noted. Appendix diagnostics  
 249 state their seed counts in captions or tables.

250 The experiments follow three questions. First, does the exact factorization explain the gradient  
 251 geometry observed in code? Second, does shunting change the path-gain field in the direction  
 252 predicted by Props. 2 and 3, and does exact path transport rescue learning when rank-1 broadcast is  
 253 insufficient? Third, do the same ingredients remain useful in capacity-calibrated supervised tasks and  
 254 across dendritic morphologies?

### 255 4.2 From Exact Factorization to Credit-Signal Fidelity

256 To test whether these performance differences correspond to better credit signals, we compare  
 257 local and backprop gradients on the same batch at the same parameter values. For each parameter  
 258 tensor  $p$ , we compute directional alignment (cosine similarity) and scale mismatch, defined as  
 259  $|\log_{10}(\|g_p^{\text{local}}\|/\|g_p^{\text{bp}}\|)|$ , aggregated by parameter count.

Across matched-weight comparisons, shunting is consistently more aligned with backprop and substantially closer in norm than the additive control: on MNIST, weighted cosine rises from 0.006 to 0.202 and scale mismatch falls from 1.053 to 0.117; on context gating, cosine rises from  $-0.007$  to 0.108 and scale mismatch falls from 2.154 to 0.036. Figure 2B rules out a trivial zero-gradient explanation: in the 5F trajectory diagnostic, additive local and backprop gradient norms remain finite throughout training, including at the final checkpoint. Figure 2C shows that the directional advantage appears throughout training. Figure 2D confirms that the theory is operational in code under the nonnegative-input setup: reconstructing gradients from Corollary 1 using exact compartment errors yields weighted relative reconstruction error below  $2 \times 10^{-7}$  and weighted scale mismatch below  $4 \times 10^{-8}$  across runs. In a few low-norm parameter blocks, cosine becomes numerically unstable even though the reconstruction itself is essentially exact, so we report the norm-based diagnostics in the figure. The only approximation made by LocalCA is therefore the broadcast substitute for  $\partial L / \partial V_n$ .

### 4.3 Mechanistic Chain: Path Gains, Broadcast Fidelity, and Oracle Transport

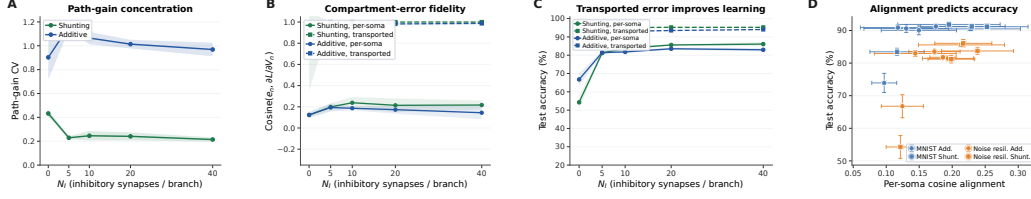
Figure 3 provides the paper’s central empirical chain: shunting compresses conductance-stage path gains, which improves direct compartment-error fidelity, which in turn makes low-bandwidth local learning effective; the transported-error oracle then isolates how much of the remaining gap is due to broadcast quality rather than other architectural differences.

Panel A measures the coefficient of variation of the exact conductance-stage path gains  $\alpha_n$  across compartments. On MNIST, shunting keeps this distribution much more compressed across inhibition levels (CV about 0.21–0.43), whereas the additive control remains much broader (CV about 0.90–1.16). Panel B then measures the sole non-local term from Corollary 1,  $\partial L / \partial V_n$ , directly. On the noise-resilience sweep, rank-1 broadcast tracks the exact compartment error somewhat better under shunting than under additive integration once inhibition is present: additive remains positively aligned throughout (about 0.14–0.19 across the sweep), while shunting rises to about 0.20–0.24 in the inhibited regimes after starting from a comparable low-inhibition baseline. The transported-error oracle obtained by recursively applying those conductance-stage path gains improves fidelity sharply in both architectures, reaching about 0.95–0.99 for additive and essentially 1.0 for shunting once inhibition is present, showing that the theorem defines an operational upper bound on low-bandwidth broadcast.

Panel C trains the same local rule using that oracle signal. On noise resilience, additive learning rises from 66.8% at  $N_I=0$  to about 81–84% across the inhibited regimes under rank-1 broadcast, and transported broadcast lifts those same conditions to about 92–94%. Shunting rises from 54.3% at  $N_I=0$  to about 81–86% under rank-1 broadcast and to about 94.3–95.2% under transported broadcast. On MNIST, transported broadcast reaches about 95.4–96.6% for additive and 96.5–97.3% for shunting. Appendix Fig. S8 reports the same ladder on flattened CIFAR-10 as a harder-data stress test: shunting rises from 28.8% under rank-1 broadcast to 47.4% under transported error, nearly matching its 49.5% backprop ceiling. We use that result diagnostically rather than as a hard-vision benchmark claim. Together, the oracle results support path-gain fidelity as the dominant bottleneck for LocalCA performance, while the smaller remaining additive–shunting gap under matched transported broadcast suggests an additional contribution from conductance-shaped local sensitivities.

Panel D collapses the chain across all conditions into a single scatter of rank-1 cosine alignment against test accuracy. Shunting cells (squares) cluster at high alignment and high accuracy; additive cells (circles) cluster at the opposite end, and the same monotonic relationship holds on both MNIST and noise resilience. This is a compact confirmation that the (path-gain compressibility  $\rightarrow$  compartment-error fidelity  $\rightarrow$  test accuracy) chain is not a dataset-specific artifact. Appendix controls fill in the boundaries: coarse quantized broadcasts preserve performance until sign-only or sparse broadcasts remove too much information (Fig. S6); rank- $K$  and path-propagation feedback close part of the transported-oracle gap (Table S10); additive normalization and DFA help but do not match the shunting conductance model (Figs. S10, S7); and explicit inhibitory cells can generate useful path-gain structure in a one-layer input-mode probe (Fig. S11, Table S8).





**Figure 3: Mechanistic chain from path gains to learning.** (A) Conductance-stage path-gain dispersion on MNIST, measured as the coefficient of variation of the exact tree attenuation  $\alpha_n$  across compartments. Shunting keeps these conductance-stage path gains much more compressed than the additive control as inhibition increases. (B) Direct compartment-error fidelity on noise resilience. Solid lines compare the rank-1 broadcast  $e_n$  to the exact compartment error  $\partial L / \partial V_n$ ; dashed lines show the transported-error oracle obtained by recursively applying conductance-stage path gains. Under the nonnegative-input setup, shunting still improves the simple rank-1 broadcast in the inhibited regimes, while the transported oracle defines the remaining approximation gap and becomes nearly exact once inhibition is present. (C) Learning with the transported-error oracle on noise resilience. Improving the broadcast field is enough to recover near-ceiling local learning, especially in the additive architecture where rank-1 broadcast fails most severely. (D) Global mechanism summary. Rank-1 cosine alignment (x-axis) vs. test accuracy (y-axis) across all conditions (additive/shunting  $\times$  five inhibition levels  $\times$  two datasets, 5 seeds each). Shunting cells (squares) cluster at high alignment and high accuracy; additive cells (circles) cluster at low alignment and low accuracy. The same monotonic chain from (A) through (C) collapses cleanly into the (alignment  $\rightarrow$  accuracy) relationship shown here, and it holds on both MNIST and noise resilience, confirming that the mechanistic chain is not a dataset-specific artifact. The broadcast-bandwidth / quantization control previously shown as panel D is now reported separately as Appendix Fig. S6.

#### 4.4 Capacity-Calibrated Competence

Having established the mechanism, we next ask whether the same conditions improve end-task learning in architectures that are already competent under backpropagation. In the capacity-calibrated regime, shunting dendritic cores remain reasonably competitive under local learning: on MNIST, Fashion-MNIST, and context gating, the best local configuration (5F, rank-1 broadcast in these architectures, local decoder) stays within about 5–6 percentage points of the matched backprop ceiling (Table S4; Fig. 4). Within the local-rule family, 5F is the strongest practical wrapper, but it remains heuristic rather than theorem-derived; Appendix Tables S1, S3, and S2 summarize that division of labor across rules and broadcast choices. Classical random-feedback baselines remain much weaker on the same task family: in the matched MNIST appendix comparison, DFA reaches 66.1% on shunting and 62.2% on additive, well below the 91.4% best shunting LocalCA result, while FA is not directly compatible with the block-structured dendritic layers (Appendix Fig. S7). This helps separate the contribution of the conductance architecture from the contribution of the LocalCA rule itself.

#### 4.5 Regime Dependence of the Shunting Advantage

The benefit of shunting varies across tasks. With matched rank-1 broadcast, shunting is modestly better on MNIST and Fashion-MNIST, and the gap grows on the inhibition-sensitive noise-resilience benchmark once inhibitory conductance is present (Fig. 4). This is the expected regime dependence: shunting helps most when input resistance is part of the useful credit geometry rather than merely an added conductance load.

#### 4.6 Morphology Shifts the Inhibitory Operating Regime

The inhibitory regime in which shunting helps most is not fixed across dendritic trees. In the morphology  $\times$  inhibition sweep (Fig. 5), depth-2 trees show their largest shunting gains near  $N_I=0$ , whereas depth-3 trees retain their strongest gains around  $N_I=5$ . This supports the broader picture that tree geometry changes how inhibitory conductance shapes effective compartment sensitivities.

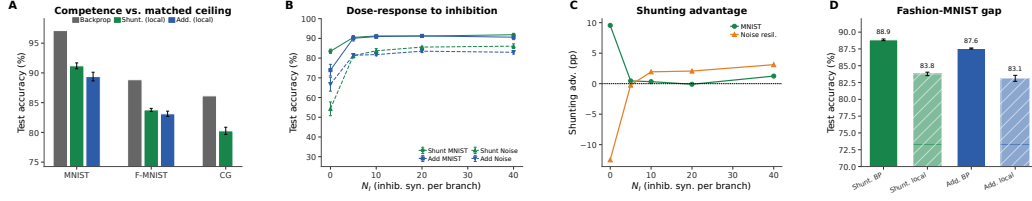


Figure 4: **Capacity-calibrated competence and regime dependence.** (A) Matched backprop ceilings vs. the best rank-1 5F LocalCA rule on MNIST, Fashion-MNIST, and context gating. (B,C) Inhibitory dose-response and shunting advantage on MNIST and noise resilience. (D) Fashion-MNIST confirms a smaller but consistent shunting advantage. Error bars denote  $\pm 1$  s.d. across seeds.

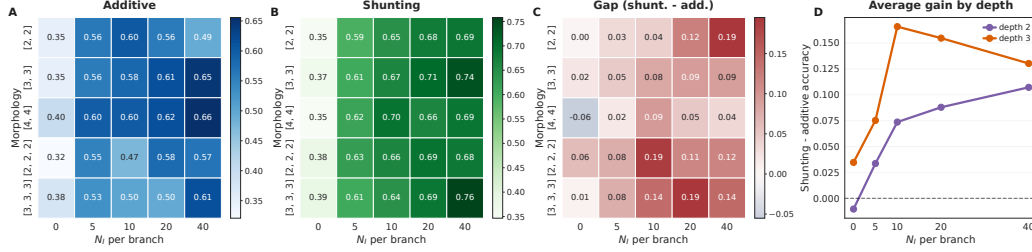


Figure 5: **Morphology shifts the useful inhibitory regime.** (A,B) Additive and shunting LocalCA across dendritic morphologies and inhibitory synapse counts. (C,D) The shunting advantage peaks at different inhibition levels for shallow and deeper trees, showing that conductance normalization and morphology interact rather than contributing independently.

**Scope.** The claim is mechanistic, not an unconstrained benchmark claim: CIFAR-10, routed feedback, and depth scaling are diagnostics of where rank-1 broadcast breaks, and shunting is not a universal solution for hard vision or biological learning.

## References

- [1] Koch, C. (1999). *Biophysics of Computation*. Oxford University Press.
- [2] Dayan, P., & Abbott, L. F. (2001). *Theoretical Neuroscience*. MIT Press.
- [3] Poirazi, P., Brannon, T., & Mel, B. W. (2003). Pyramidal neuron as two-layer neural network. *Neuron*, 37(6), 989–999.
- [4] London, M., & Häusser, M. (2005). Dendritic computation. *Annual Review of Neuroscience*, 28, 503–532.
- [5] Urbanczik, R., & Senn, W. (2014). Learning by the dendritic prediction of somatic spiking. *Neuron*, 81(3), 521–528.
- [6] Carandini, M., & Heeger, D. J. (2012). Normalization as a canonical neural computation. *Nature Reviews Neuroscience*, 13(1), 51–62.
- [7] Holt, G. R., & Koch, C. (1997). Shunting inhibition does not have a divisive effect on firing rates. *Neural Computation*, 9(5), 1001–1013.
- [8] Vogels, T. P., Sprekeler, H., Zenke, F., Clopath, C., & Gerstner, W. (2011). Inhibitory plasticity balances excitation and inhibition in sensory pathways and memory networks. *Science*, 334(6062), 1569–1573.
- [9] Lillicrap, T. P., Cownden, D., Tweed, D. B., & Akerman, C. J. (2016). Random synaptic feedback weights support error backpropagation for deep learning. *Nature Communications*, 7, 13276.

- [10] Nøkland, A. (2016). Direct feedback alignment provides learning in deep neural networks. *NeurIPS*, 29.
- [11] Guerguiev, J., Lillicrap, T. P., & Richards, B. A. (2017). Towards deep learning with segregated dendrites. *eLife*, 6, e22901.
- [12] Sacramento, J., Costa, R. P., Bengio, Y., & Senn, W. (2018). Dendritic cortical microcircuits approximate the backpropagation algorithm. *NeurIPS*, 31.
- [13] Whittington, J. C., & Bogacz, R. (2019). Theories of error back-propagation in the brain. *Trends in Cognitive Sciences*, 23(3), 235–250.
- [14] Richards, B. A., & Lillicrap, T. P. (2019). Dendritic solutions to the credit assignment problem. *Current Opinion in Neurobiology*, 54, 28–36.
- [15] Scellier, B., & Bengio, Y. (2017). Equilibrium propagation. *Frontiers in Computational Neuroscience*, 11, 24.
- [16] Song, Y., Millidge, B., Salvatori, T., Lukasiewicz, T., Xu, Z., & Bogacz, R. (2024). Inferring neural activity before plasticity as a foundation for learning beyond backpropagation. *Nature Neuroscience*, 27, 348–358.
- [17] Gretton, A., Bousquet, O., Smola, A. J., & Schölkopf, B. (2005). Measuring statistical dependence with Hilbert-Schmidt norms. *Algorithmic Learning Theory*, 63–77.
- [18] Frémaux, N., & Gerstner, W. (2016). Neuromodulated spike-timing-dependent plasticity, and theory of three-factor learning rules. *Frontiers in Neural Circuits*, 9, 85.
- [19] Bellec, G., et al. (2020). A solution to the learning dilemma for recurrent networks of spiking neurons. *Nature Communications*, 11, 3625.
- [20] Welford, B. P. (1962). Note on a method for calculating corrected sums of squares and products. *Technometrics*, 4(3), 419–420.
- [21] Turrigiano, G. G. (2008). The self-tuning neuron: synaptic scaling of excitatory synapses. *Cell*, 135(3), 422–435.
- [22] Payeur, A., Guerguiev, J., Zenke, F., Richards, B. A., & Naud, R. (2021). Burst-dependent synaptic plasticity can coordinate learning in hierarchical circuits. *Nature Neuroscience*, 24(7), 1010–1019.
- [23] Greedy, W., Zhu, H. W., Pemberton, J., Mellor, J., & Ponte Costa, R. (2022). Single-phase deep learning in cortico-cortical networks. *NeurIPS*, 35, 24213–24225.
- [24] Haider, P., Ellenberger, B., Kriener, L., Jordan, J., Senn, W., & Petrovici, M. A. (2021). Latent Equilibrium: A unified learning theory for arbitrarily fast computation with arbitrarily slow neurons. *NeurIPS*, 34.
- [25] Hinton, G. (2022). The forward-forward algorithm. *arXiv:2212.13345*.
- [26] Dellaferrera, G., & Kreiman, G. (2022). Error-driven input modulation: Solving the credit assignment problem without a backward pass. *Proceedings of the 39th International Conference on Machine Learning, Proceedings of Machine Learning Research*, 162, 4937–4955.
- [27] Lee, D.-H., et al. (2015). Difference target propagation. *ECML*, 498–515.
- [28] Meulemans, A., et al. (2021). Credit assignment in neural networks through deep feedback control. *NeurIPS*, 34.
- [29] Millidge, B., Seth, A. K., & Buckley, C. L. (2021). Predictive coding: A theoretical and experimental review. *arXiv:2107.12979*.
- [30] Koch, C., Poggio, T., & Torre, V. (1983). Nonlinear interactions in a dendritic tree. *PNAS*, 80(9), 2799–2802.

- [31] Silver, R. A. (2010). Neuronal arithmetic. *Nature Reviews Neuroscience*, 11(7), 474–489.
- [32] Max, K., Kriener, L., Pineda García, G. et al. (2024). Learning efficient backprojections across cortical hierarchies in real time. *Nature Machine Intelligence*, 6(6), 619–630.
- [33] Ma, C., Wu, J., Si, C., & Tan, K. C. (2024). Scaling supervised local learning with augmented auxiliary networks. *International Conference on Learning Representations (ICLR)*.
- [34] Lv, C., Xu, J., Lu, Y., Wang, X., Wang, Z., Xu, Z., Yu, D., Du, X., Zheng, X., & Huang, X. (2025). Dendritic Localized Learning: Toward Biologically Plausible Algorithm. *Proceedings of the 42nd International Conference on Machine Learning, Proceedings of Machine Learning Research*, 267, 41682–41700.
- [35] Erdogan, M., Pehlevan, C., & Erdogan, A. T. (2025). Error broadcast and decorrelation as a potential artificial and natural learning mechanism. *OpenReview*, NeurIPS 2025 spotlight.
- [36] Kao, C.-H., & Hariharan, B. (2024). Counter-Current Learning: A Biologically Plausible Dual Network Approach for Deep Learning. *Advances in Neural Information Processing Systems*, 37.
- [37] Xiao, H., Rasul, K., & Vollgraf, R. (2017). Fashion-MNIST: A novel image dataset for benchmarking machine learning algorithms. *arXiv:1708.07747*.
- [38] Song, S., Sjöström, P. J., Reigl, M., Nelson, S., & Chklovskii, D. B. (2005). Highly nonrandom features of synaptic connectivity in local cortical circuits. *PLoS Biology*, 3(3), e68.
- [39] Buzsáki, G., & Mizuseki, K. (2014). The log-dynamic brain: How skewed distributions affect network operations. *Nature Reviews Neuroscience*, 15(4), 264–278.
- [40] The MICrONS Consortium (2025). Functional connectomics spanning multiple areas of mouse visual cortex. *Nature*, 640(8058), 435–447.
- [41] Loewenstein, Y., Kuras, A., & Rumpel, S. (2011). Multiplicative Dynamics Underlie the Emergence of the Log-Normal Distribution of Spine Sizes in the Neocortex In Vivo. *Journal of Neuroscience*, 31(26), 9481–9488.
- [42] Megías, M., et al. (2001). Total number and distribution of inhibitory and excitatory synapses on hippocampal CA1 pyramidal cells. *Neuroscience*, 102(3), 527–540.
- [43] Francioni, V., Tang, V. D., Toloza, E. H. S. et al. (2026). Vectorized instructive signals in cortical dendrites. *Nature*. <https://doi.org/10.1038/s41586-026-10190-7>
- [44] Iyer, A., Grewal, K., Velu, A., Souza, L. O., Forest, J., & Ahmad, S. (2022). Avoiding catastrophe: Active dendrites enable multi-task learning in dynamic environments. *Frontiers in Neurobotics*, 16, 846219.

## A Supplementary Results

### Mechanistic Support and Broadcast Controls

#### Rule-Family Ranking

#### Which Components Carry?

| Dataset        | Rule | Top-10 valid | Top-10 test  |
|----------------|------|--------------|--------------|
| MNIST          | 3F   | 0.611        | 0.622        |
| MNIST          | 4F   | 0.620        | 0.628        |
| MNIST          | 5F   | <b>0.912</b> | <b>0.916</b> |
| Context gating | 3F   | 0.398        | 0.396        |
| Context gating | 4F   | 0.411        | 0.411        |
| Context gating | 5F   | <b>0.807</b> | <b>0.789</b> |

Table S1: **Rule-family ranking.** Top-10 mean across completed local-competence sweeps.

| Component                | Representative effect                                                                                                                                                                                                                                                                                                 | Interpretation                                                                                                                                                                                                                                                                                                                               |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Shunting architecture    | In the nonnegative-input noise-resilience sweep, additive performs better at $N_I=0$ (0.668 vs. 0.543), but shunting overtakes it once inhibition is present (0.837 vs. 0.818 at $N_I=10$ , 0.861 vs. 0.830 at $N_I=40$ ).                                                                                            | The shunting gain is operating-regime dependent rather than uniform: it appears once inhibition shapes conductance scaling strongly enough to help the shared broadcast.                                                                                                                                                                     |
| Broadcast quality        | On MNIST (5F), shunting per-soma reaches $0.912 \pm 0.005$ , while local-mismatch remains at $0.146 \pm 0.046$ .                                                                                                                                                                                                      | Correctly matching the broadcast field to the somatic error matters far more than arbitrary local mismatch heuristics.                                                                                                                                                                                                                       |
| Transported oracle       | Across inhibited noise-resilience regimes, additive local learning rises from roughly 81–84% under per-soma to 92–94% under transported broadcast; shunting rises from roughly 81–86% to 95–95.2%.                                                                                                                    | The oracle remains the strongest causal evidence that broadcast fidelity, not just architecture, is the dominant bottleneck.                                                                                                                                                                                                                 |
| Rule hierarchy           | In the competence sweeps, MNIST improves from 0.622 (3F) and 0.628 (4F) to 0.916 (5F); context gating improves from 0.396/0.411 to 0.789.                                                                                                                                                                             | 5F is the strongest practical rule in this paper, though its extra factor remains heuristic rather than theorem-derived.                                                                                                                                                                                                                     |
| 5F stability             | Tightening the $\phi$ clamp to $[0.5, 2.0]$ changes MNIST by only about +0.4 pp; varying EMA from 0.05 to 0.20 changes it by less than 0.2 pp.                                                                                                                                                                        | The practical gain from 5F is not especially brittle to its main tuning knobs.                                                                                                                                                                                                                                                               |
| Structured feedback      | In the cue-routing rank sweep, learned-routing additive local learning remains near ceiling while learned-routing shunting rises from 0.742 at random rank-1 to 0.846 at random rank-2; the current tuned pathway-vector variant reaches 0.815.                                                                       | Routed tasks remain the main failure mode of rank-1 feedback, but the robust conclusion is that higher rank helps; the best structured fix is not yet settled enough to anchor a headline claim.                                                                                                                                             |
| Random low-rank channels | On the shunting noise-resilience bridge, random low-rank broadcast improves from 0.453 at $K=1$ to 0.764 at $K=4$ and 0.795 at $K=8$ , while exact path transport reaches 0.847; on cue routing, the best unstructured low-rank setting is $K=2$ at 0.846.                                                            | Higher-rank unstructured broadcast now closes a large fraction of the rank-1 gap, but it still does not recover the transported-oracle ceiling and does not by itself establish a structure-specific advantage.                                                                                                                              |
| Path propagation proxy   | In the shunting noise-resilience bridge, the current morphology-aware path-propagation proxy reaches $0.762 \pm 0.047$ , well above per-soma ( $0.662 \pm 0.016$ ) and close to random low-rank $K=4$ ( $0.764 \pm 0.025$ ), though still below exact path transport ( $0.847 \pm 0.018$ ).                           | The present recursive attenuation proxy is a meaningful constructive baseline: it closes much of the oracle gap, though a stronger strictly local approximation to transported error remains open.                                                                                                                                           |
| Morphology regime        | In the exploratory noise-resilience regime map (Fig. 5), depth-2 trees show their largest shunting gains near $N_I = 0$ , whereas depth-3 trees retain their strongest gains around $N_I = 5$ .                                                                                                                       | Morphology appears to shift the inhibitory operating regime in which shunting is most useful, suggesting that conductance normalization and tree geometry interact rather than contributing independently.                                                                                                                                   |
| Activation choice        | In the five-seed frozen activation audit, identity outperforms frozen <code>param_tanh</code> for LocalCA on noise resilience in both architectures (0.590/0.859 vs. 0.109/0.772 for additive at $N_I=0/10$ ; 0.411/0.769 vs. 0.331/0.701 for shunting), while the backprop effect remains strongly regime-dependent. | The post-voltage firing-rate transform is a real modeling variable, not just an implementation detail. Even a fixed saturating nonlinearity changes the operating point substantially, so the main paper’s mechanistic claims should be read in the learned bounded <code>param_tanh</code> regime used throughout the headline experiments. |

Table S2: **Which components carry the observed gains?** A compact summary of the empirical division of labor across architecture, broadcast design, and LocalCA rule variants.

| Core     | Broadcast      | Decoder  | Test (mean $\pm$ std) |
|----------|----------------|----------|-----------------------|
| Shunting | per-soma       | local    | 0.912 $\pm$ 0.005     |
| Shunting | per-soma       | backprop | 0.909 $\pm$ 0.008     |
| Shunting | local-mismatch | local    | 0.146 $\pm$ 0.046     |
| Shunting | local-mismatch | backprop | 0.146 $\pm$ 0.037     |
| Additive | per-soma       | local    | 0.894 $\pm$ 0.007     |
| Additive | per-soma       | backprop | 0.900 $\pm$ 0.001     |
| Additive | local-mismatch | local    | 0.342 $\pm$ 0.058     |
| Additive | local-mismatch | backprop | 0.348 $\pm$ 0.095     |

Table S3: **Local-mismatch recheck (MNIST, 5F)**. Per-soma is consistently strong; local-mismatch remains much weaker.

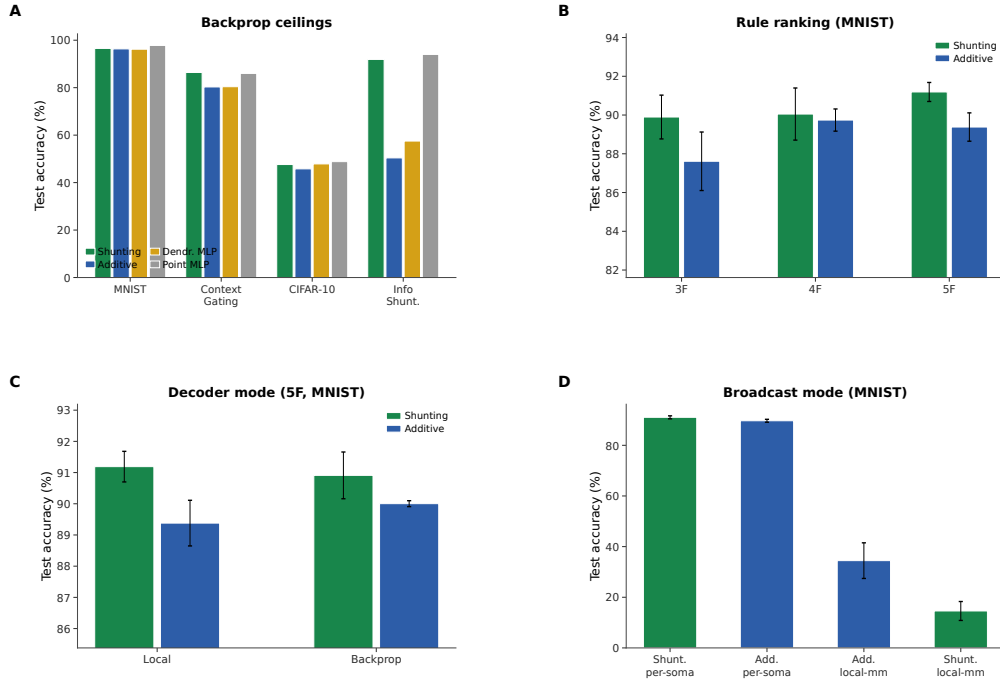


Figure S1: **Capacity calibration (supplementary)**. (A) Phase 1 backprop ceilings across all architectures and datasets. (B) Rule-family ranking: 5F consistently outperforms 4F and 3F. (C) Decoder mode comparison (local vs. backprop decoder, 5F MNIST). (D) Broadcast mode comparison: per-soma is required for strong performance; local-mismatch fails.

| Dataset        | BP ceiling | Best local (5F)   | Gap  |
|----------------|------------|-------------------|------|
| MNIST          | 0.971      | $0.914 \pm 0.003$ | 5.7% |
| Fashion-MNIST  | 0.889      | $0.838 \pm 0.003$ | 5.1% |
| Context gating | 0.861      | $0.803 \pm 0.006$ | 5.8% |

Table S4: **Local competence.** Backprop ceilings are matched-capacity shunting references from dedicated 5-seed standard-training refreshes for MNIST and context gating and from the dedicated 5-seed Fashion-MNIST competence sweep. Local values are 5F with per-soma broadcast on shunting dendritic cores. Context gating additionally uses an HSIC auxiliary objective (weight 0.01; Appendix E). Error bars on local values:  $\pm 1$  s.d. across 5 seeds.

| Condition                                               | Metric           | Value                   |
|---------------------------------------------------------|------------------|-------------------------|
| Decoder: local vs backprop vs frozen (MNIST, toy model) | test acc         | 0.379 vs 0.379 vs 0.176 |
| Shunting vs additive, matched (MNIST, toy model)        | $\Delta$ test    | +0.210                  |
| Per-soma, path on vs off (synthetic task, toy model)    | test / MI(E,I;C) | -0.003 / +0.017         |

Table S5: **Mechanistic ablations.** Results from controlled small-network experiments using minimal architectures to isolate individual factors. These runs use a small 2-layer dendritic network (branch factor [9]) trained for 30 epochs; they are not directly comparable to the headline capacity-calibrated results.

#### 438 Broadcast Mode Comparison and Local-Mismatch Recheck

#### 439 Competence, Verification, and Stress Tests

#### 440 Phase 1 Capacity Ceilings

#### 441 Local Competence and Regime Dependence

#### 442 Ablation Results

#### 443 Extended Gradient Analysis and $N_I$ Sweep Detail

#### 444 Verification and Reproducibility

#### 445 Additional Stress Tests

#### 446 Broadcast Bandwidth (Quantization) Control

#### 447 FA/DFA Baseline Comparison

#### 448 CIFAR-10 Results

449 Table S6 provides the exact values for the appendix CIFAR-10 figure. In this stronger compact direct-  
450 I-stream family, shunting performs well under standard training (49.5% vs. 46.6% for additive), so  
451 the harder-data stress test does not unfairly penalize the shunting architecture. The broadcast-fidelity  
452 ladder also survives once decoder-aware soma mapping is used: additive LocalCA rises from 32.6%  
453 under rank-1 per-soma broadcast to 42.3% under transported error, while shunting rises from 28.8%  
454 under per-soma to 34.7% with random low-rank  $K=4$  and 47.4% with transported error, nearly  
455 matching the 49.5% shunting backprop ceiling.

456 **CIFAR-10 soma-on extension.** We also tested the soma-on extension on the same com-  
457 pact direct-I-stream depth-4 family with quantile init (Fig. S9; Table S7; 3 seeds). With  
458 `somatic_synapses=true`, path-transported LocalCA stays strong in both cores: additive reaches  
459  $45.5 \pm 0.6\%$  and shunting reaches  $46.6 \pm 1.0\%$ . Relative to the soma-off results, the additive  
460 path-transport condition improves from 42.3% to 45.5%, while shunting remains within 1 pp of its  
461 soma-off path-transport ceiling (47.4%) and within 3 pp of the matched shunting backprop ceiling  
462 (49.5%). Per-soma rank-1 broadcast also becomes much more competitive for additive ( $41.6 \pm 0.6\%$ )  
463 than for shunting ( $35.9 \pm 1.2\%$ ), while low-rank  $K=4$  remains weak in both cores. Figure S9 makes

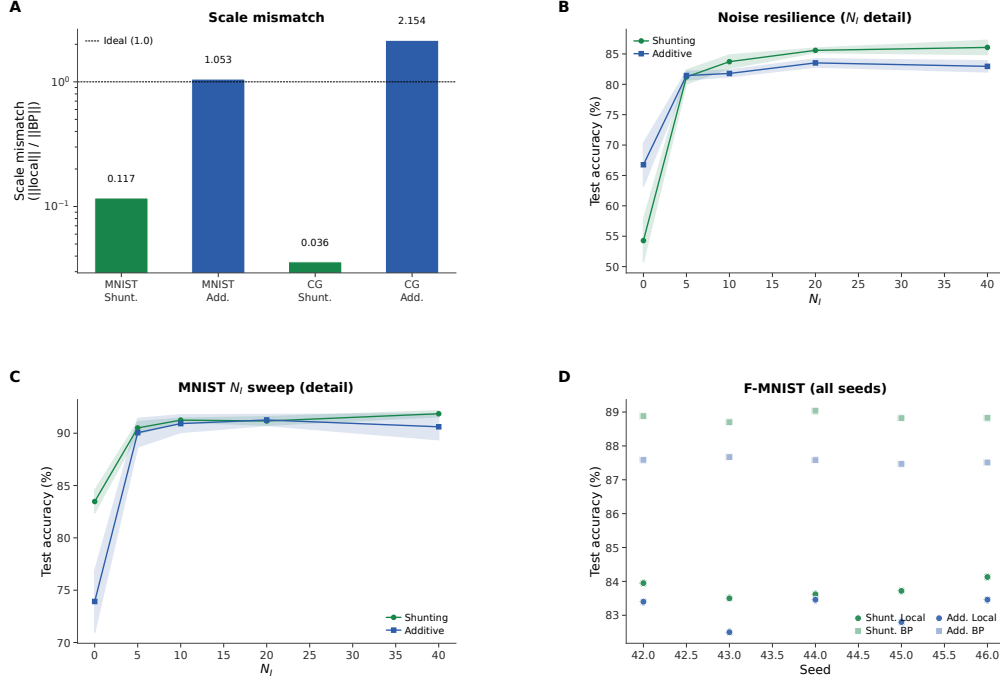


Figure S2: **Extended gradient and  $N_I$  analysis (supplementary).** (A) Scale mismatch bars: shunting achieves near-ideal scale (0.117); additive exhibits order-of-magnitude distortion ( $> 1.0$ ). (B) Noise-resilience  $N_I$  dose-response with error bands ( $\pm 1$  s.d.). (C) MNIST  $N_I$  dose-response detail with error bands. (D) Fashion-MNIST individual seeds for all conditions, showing consistency across runs.

| Condition              | Model    | Test accuracy                    |
|------------------------|----------|----------------------------------|
| standard               | additive | $46.6 \pm 1.5$                   |
| standard               | shunting | <b><math>49.5 \pm 0.7</math></b> |
| LocalCA per-soma       | additive | $32.6 \pm 0.6$                   |
| LocalCA per-soma       | shunting | $28.8 \pm 0.8$                   |
| LocalCA path transport | additive | $42.3 \pm 1.2$                   |
| LocalCA path transport | shunting | <b><math>47.4 \pm 0.6</math></b> |
| shunting low-rank      | $K=4$    | $34.7 \pm 1.1$                   |

Table S6: **Exact CIFAR-10 values for the compact mechanism extension (5 seeds, validation-best epoch).** This table gives the exact numbers for the appendix harder-data ladder in Fig. S8. The key technical detail is that LocalCA maps output error back to decoder-input/soma space through the decoder Jacobian when the decoder is nonlinear. With decoder-aware error mapping, transported error again acts as a strong upper bound, nearly matching the stronger shunting backprop ceiling, while naive rank-1 per-soma broadcast still underperforms and random low-rank feedback closes part of that gap.

464 the two useful views explicit: the absolute soma-on ordering still favors path transport strongly, and  
465 the matched soma-off to soma-on gain is largest for additive per-soma while shunting transported  
466 error is already near its soma-off ceiling. On this harder dataset, the soma channel clearly helps, but  
467 it still does not eliminate the importance of broadcast quality.



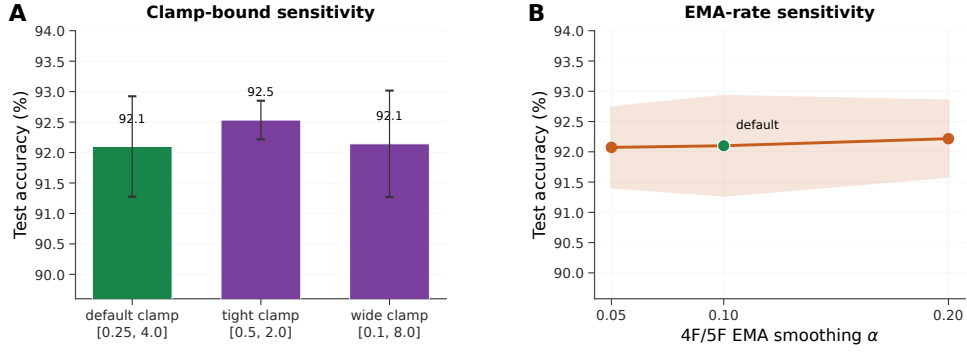


Figure S3: **5F sensitivity analysis (MNIST, shunting, per-soma broadcast)**. (A) Sensitivity to the  $\phi_n$  clamp interval. The default clamp [0.25, 4.0] is already strong ( $92.1 \pm 0.8\%$ ); tightening it to [0.5, 2.0] slightly improves performance ( $92.5 \pm 0.3\%$ ), while widening it to [0.1, 8.0] is effectively neutral ( $92.1 \pm 0.9\%$ ). (B) Sensitivity to the 4F/5F EMA rate used for online statistics. Varying  $\alpha$  from 0.05 to 0.20 changes mean test accuracy by less than 0.2 pp. Together these results suggest that the 5F gain is not driven by a brittle single hyperparameter choice, even though the factor itself remains heuristic. Errors:  $\pm 1$  s.d. across 3 seeds.

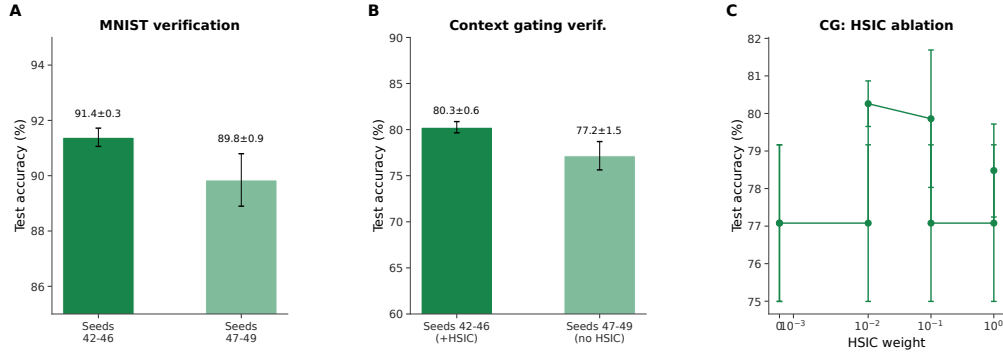


Figure S4: **Verification and reproducibility (supplementary)**. (A) MNIST verification: main seeds (42–46) yield  $91.4 \pm 0.3\%$ ; held-out seeds (47–49) yield  $89.8 \pm 0.9\%$ , confirming generalization. (B) Context gating verification: main seeds (+HSIC)  $80.3 \pm 0.6\%$ ; held-out seeds (no HSIC)  $77.2 \pm 1.5\%$ . The roughly 3 pp gap reflects HSIC removal, not seed sensitivity. (C) HSIC weight ablation on context gating: moderate weights (0.01–0.1) perform best.

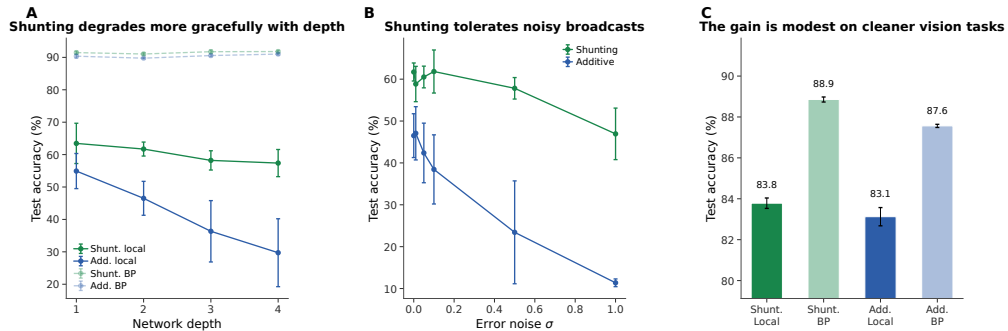


Figure S5: **Additional stress tests**. (A) Depth scaling: shunting local (green) degrades more gracefully from 63.5% to 57.4%; additive local (blue) drops from 54.9% to 29.7%. Backprop ceilings (dashed) remain stable. (B) Broadcast-noise robustness: shunting remains stable for  $\sigma \leq 0.1$ ; additive degrades rapidly and reaches chance at  $\sigma = 1.0$ . (C) Fashion-MNIST: the shunting gain is modest on this cleaner benchmark (83.8% local vs. 88.9% backprop), consistent with the regime-dependent story in the main text.

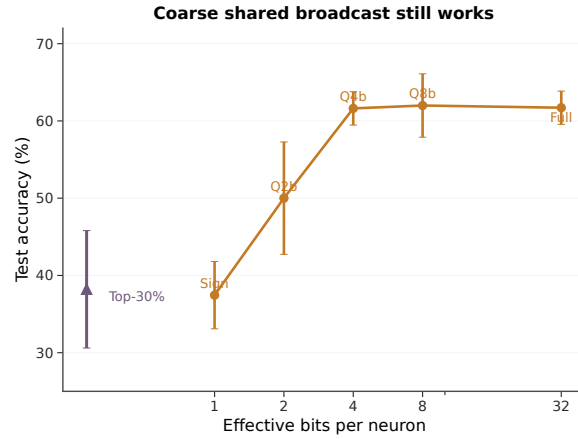


Figure S6: **Coarse shared broadcast still works.** Test accuracy as a function of effective broadcast bandwidth. Full-precision, 8-bit, 4-bit, and 2-bit quantization of the shared error signal all reach essentially the same accuracy (61–62%;  $p > 0.9$  for 4-bit vs. full), while a sign-only broadcast and a sparse top-30% broadcast degrade sharply. This shows that the shunting LocalCA signal carries most of its useful information in its direction rather than in fine-grained magnitude, consistent with the mechanism-summary picture in the main text. This panel is reported separately so the main figure can carry the full causal chain (path-gain compressibility  $\rightarrow$  alignment  $\rightarrow$  oracle  $\rightarrow$  global summary).

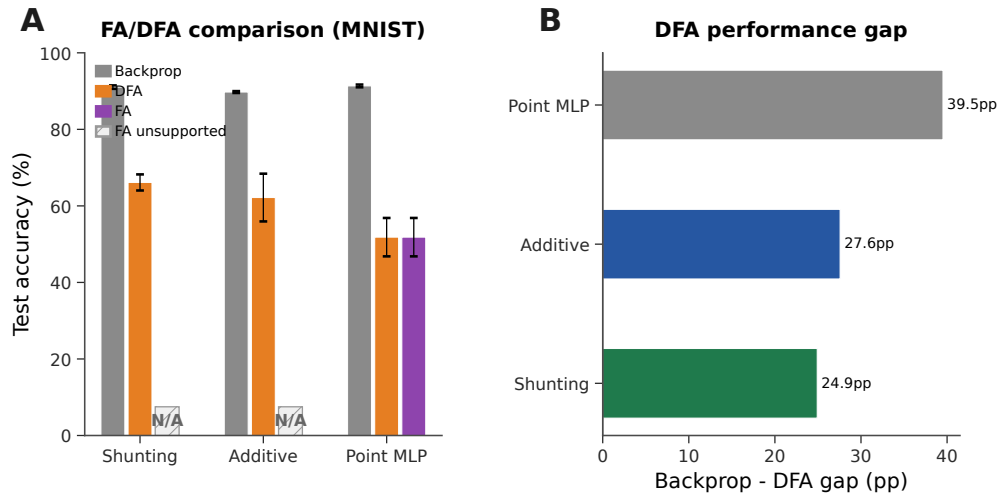


Figure S7: **Feedback alignment baselines (MNIST).** (A) Grouped comparison: standard backprop (neutral gray), DFA (orange), and FA (purple). Hatched N/A markers indicate that FA is not implemented for the block-structured dendritic architectures because the random feedback matrices are not dimensionally compatible with the tree-structured branch updates. DFA achieves 66.1% on shunting vs. 62.2% additive vs. 51.8% point MLP. (B) Backprop–DFA gap: dendritic architectures (24.9–27.6 pp) show smaller gaps than point MLPs (39.5 pp), suggesting conductance-based architecture is partially compatible with random feedback.

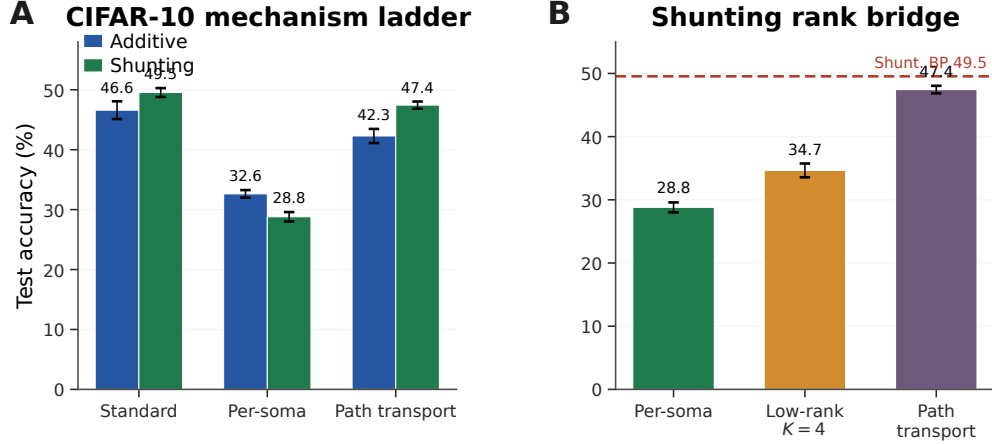


Figure S8: **CIFAR-10 mechanism stress test in the compact direct-I-stream family.** (A) Matched additive vs. shunting ladder with a nonlinear decoder and decoder-aware soma-error mapping. Shunting is competitive under standard training ( $49.5 \pm 0.7\%$ ) and transported-error LocalCA nearly recovers that ceiling ( $47.4 \pm 0.6\%$ ), while naive rank-1 broadcast remains much weaker. (B) Shunting rank bridge on the same family. Random low-rank feedback improves over rank-1 broadcast, but exact transported error remains the strongest signal. This figure is a harder-data diagnostic, not a claim of competitive CIFAR-10 benchmarking.

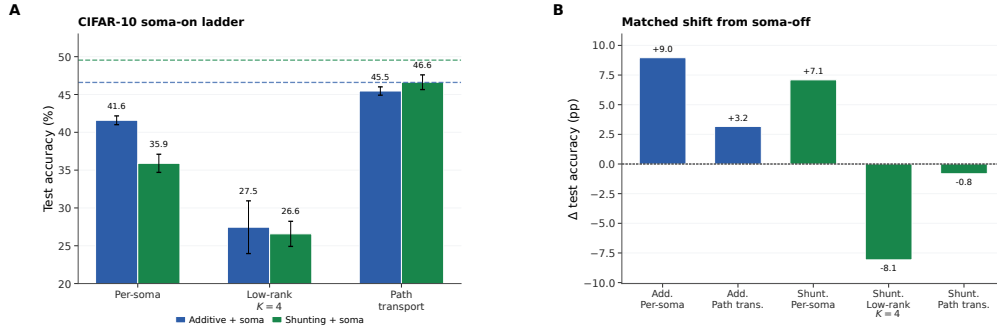


Figure S9: **CIFAR-10 compact direct-I-stream depth-4 with `somatic_synapses=true`.** (A) Absolute soma-on ladder on the compact direct-I-stream depth-4 family. Path transport stays strongest in both cores and remains near the matched backprop ceilings (dashed lines), while low-rank  $K=4$  remains weak. (B) Matched shift from soma-off to soma-on for the conditions with a clean paired baseline. Soma helps additive strongly under per-soma and path transport, improves shunting per-soma, leaves shunting transported error close to its soma-off value, and worsens shunting low-rank  $K=4$ . Additive low-rank  $K=4$  is omitted from panel B because the soma-off mechanism package did not include that exact matched condition.

| Condition (soma-on)       | Additive       | Shunting                         |
|---------------------------|----------------|----------------------------------|
| LocalCA per-soma (rank-1) | $41.6 \pm 0.6$ | $35.9 \pm 1.2$                   |
| LocalCA low-rank $K=4$    | $27.5 \pm 3.5$ | $26.6 \pm 1.7$                   |
| LocalCA path transport    | $45.5 \pm 0.6$ | <b><math>46.6 \pm 1.0</math></b> |
| soma-off path transport   | $42.3 \pm 1.2$ | $47.4 \pm 0.6$                   |
| standard BP (ceiling)     | $46.6 \pm 1.5$ | $49.5 \pm 0.7$                   |

Table S7: **CIFAR-10 compact direct-I-stream depth-4 with `somatic_synapses=true`.** Adding direct somatic synapses and using quantile reactivation init improves the additive transported-error condition by about 3.2 pp and keeps the shunting transported-error condition within 1 pp of its soma-off counterpart. Per-soma (rank-1) becomes much more competitive for additive than for shunting, while low-rank  $K=4$  remains weak in both cores. The soma channel therefore helps, but does not remove the need for a strong credit signal.

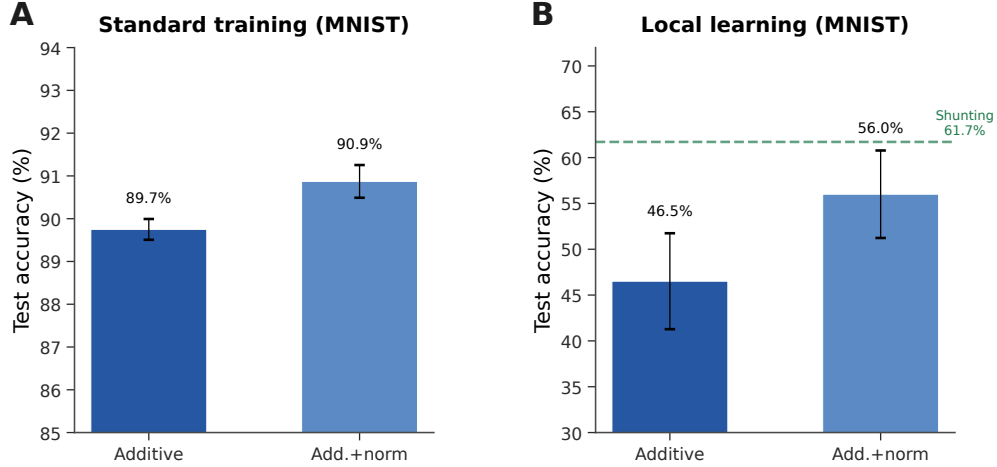


Figure S10: **Additive + normalization control (MNIST).** (A) Standard backprop: normalization provides a small boost (89.7%  $\rightarrow$  90.9%). (B) Local learning: normalization improves additive from 46.5% to 56.0% (+9.5 pp), partially closing the gap to shunting (61.7%, dashed green). The remaining gap ( $-5.7$  pp) indicates that shunting provides benefits beyond simple voltage normalization.

#### Additive Normalization Control

#### Structured Feedback, Activation, and Supportive Extensions

#### Morphology $\times$ Inhibition Regime Map

The main-text morphology figure shows the supportive regime map, while the appendix keeps the summary tables and architecture details that go with it. We treat it as supportive evidence that morphology changes the inhibitory operating regime in which shunting is most useful; a fuller mechanistic account would still require matched theory diagnostics for each cell in the grid.

#### Input-Mode Probe: Direct Inhibitory Stream vs. Explicit Inhibitory Cells

The headline experiments use `input_mode=1`, where the transfer layer provides nonnegative excitatory and inhibitory streams from the external input and the excitatory dendrites receive learned I-to-E conductance banks directly. To test whether the same mechanism can be generated by an explicit feedforward inhibitory population, we ran a one-feedforward-layer noise-resilience probe that preserves the relevant dendritic depth ([3, 3] morphology) while removing an unnecessary second feedforward population. In this clean setting, explicit inhibitory cells reach  $94.2 \pm 0.2\%$  under path-transport LocalCA when their dendrites use the same update policy as excitatory dendrites (`explicit_inhibitory_update_mode=local_ca`),  $93.8 \pm 0.2\%$  when those inhibitory-cell updates are frozen (`freeze`), and  $95.1 \pm 0.3\%$  under standard backprop. Direct input-driven inhibition gives the expected broadcast-fidelity ladder on the same one-layer architecture: per-soma rank-1 reaches  $85.2 \pm 0.7\%$ , while exact path transport reaches  $95.2 \pm 0.0\%$ . Thus explicit inhibitory cells can generate useful path-gain structure once the architecture isolates the dendritic-path question. The older two-feedforward-layer explicit-I diagnostic was much weaker (about 58–63% under LocalCA despite a 94.7% backprop ceiling), so we treat it as evidence that deeper explicit  $E \rightarrow I \rightarrow E$  graphs need graph-aware local errors, not as evidence against inhibition-mediated path gains.

**Fitted inhibitory path-gain diagnostic.** We also asked whether the direct-I-stream models actually use inhibitory synapses in the way implied by Prop. 2. We analyzed the five successful fixed-router shunting soma-on cue-routing LocalCA runs and measured first-layer inhibitory synaptic activation, branch-level local input resistance, and a pathway-inhibition balance statistic. The sign of the branch balance is not fixed across random seeds, so the meaningful quantity is sign-invariant: whether the balance carries context information, not whether branch 1 or branch 2 always encodes a named cue. Table S9 shows that the pathway-inhibition balance decodes context well after allowing this seed-

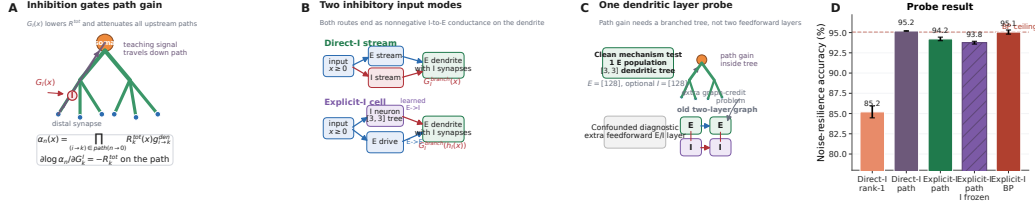


Figure S11: **Pathwise inhibition story and one-layer input-mode probe.** (A) In a branched conductance tree, the error path gain  $\alpha_n$  is the product of local input resistances and dendritic coupling conductances along the path to the soma. Inhibitory conductance on any compartment along that path lowers local input resistance and attenuates all upstream credit paths. (B) The same branch-level inhibitory conductance can be supplied directly from a nonnegative inhibitory input stream (`input_mode=1`) or through an explicit feedforward inhibitory cell (`input_mode=0`). (C) The clean mechanism probe keeps the dendritic tree depth ([3, 3]) but uses only one feedforward dendritic population; adding a second feedforward E/I layer tests a deeper graph-credit problem rather than the path-gain mechanism itself. (D) Noise-resilience results for the one-layer probe. Direct-I path transport reaches the explicit-I backprop ceiling, and explicit inhibitory cells with path-transport LocalCA also remain near ceiling. This supports the interpretation that explicit inhibitory cells can generate useful path-gain structure when the architecture isolates the dendritic-path question.

| Inhibitory input mode                                  | Training / broadcast                           | Test accuracy                       |
|--------------------------------------------------------|------------------------------------------------|-------------------------------------|
| Direct inhibitory stream ( <code>input_mode=1</code> ) | LocalCA, per-soma rank-1                       | $0.852 \pm 0.007$                   |
| Direct inhibitory stream ( <code>input_mode=1</code> ) | LocalCA, exact path transport                  | <b><math>0.952 \pm 0.000</math></b> |
| Explicit I cells ( <code>input_mode=0</code> )         | LocalCA, path transport, I-cell updates on     | $0.942 \pm 0.002$                   |
| Explicit I cells ( <code>input_mode=0</code> )         | LocalCA, path transport, I-cell updates frozen | $0.938 \pm 0.002$                   |
| Explicit I cells ( <code>input_mode=0</code> )         | standard backprop                              | $0.951 \pm 0.003$                   |

Table S8: **One-feedforward-layer input-mode path-gain probe on noise resilience (3 seeds).** All rows use one excitatory dendritic population with [3, 3] morphology and nonnegative inputs. The direct-I rows have no explicit inhibitory neurons; the input stream drives learned branch-level inhibitory conductance banks directly. The explicit-I rows use a matched inhibitory population with [3, 3] morphology. `local_ca` updates explicit inhibitory-cell dendrites with the same policy used for excitatory dendrites, while `freeze` holds those dendrites fixed but still lets excitatory I-to-E synapses learn. Exact path transport closes the direct-I rank-1 gap, and explicit inhibitory cells nearly match both direct-I path transport and the explicit-I backprop ceiling.

498 dependent polarity, and that inhibitory conductance is negatively correlated with local input resistance.  
499 This does not by itself prove an optimized interneuron circuit, but it supports the mechanistic link  
500 used throughout the paper: learned inhibitory conductance can create input-dependent path-gain  
501 differences by modulating branch resistance.

## 502 Cue-Routing Feedback-Rank Diagnostic

503 Cue routing tests the regime where one shared teaching signal is expected to fail: context determines  
504 which noisy cue is reliable, so different pathways should receive different credit. Figure S12 shows  
505 that the benchmark is learnable and that moving beyond rank-1 feedback helps, but the current  
506 path-structured variant does not yet beat the best unstructured low-rank setting. We therefore interpret  
507 this result as a failure-mode and rank-requirement diagnostic, not as a solved structured-feedback  
508 method.

| Diagnostic                                                           | Mean $\pm$ s.d.        | Interpretation                                                     |
|----------------------------------------------------------------------|------------------------|--------------------------------------------------------------------|
| $I(\text{context}; \text{pathway-inhibition balance})$               | $0.685 \pm 0.209$ bits | Context is visible in the inhibitory balance.                      |
| Best-polarity context decoding from inhibitory balance               | $0.796 \pm 0.109$      | Branch polarity varies by seed, but the balance is informative.    |
| $\text{Corr}(\text{inhibitory conductance, local input resistance})$ | $-0.424 \pm 0.122$     | More inhibition lowers local resistance, as predicted by shunting. |

Table S9: **Fitted direct-I cue-routing diagnostic (5 successful fixed-router shunting soma-on LocalCA runs).** In `input_mode=1`, inhibitory synapse banks receive a nonnegative input stream directly. In trained cue-routing models, the balance of inhibitory activation across routed pathways carries context information and higher inhibitory conductance coincides with lower local input resistance. Because the sign of the pathway balance can flip across seeds, the decoding result is reported with best polarity and should be read as evidence for context-dependent inhibitory path gating, not as a fixed anatomical assignment of one branch to one cue.

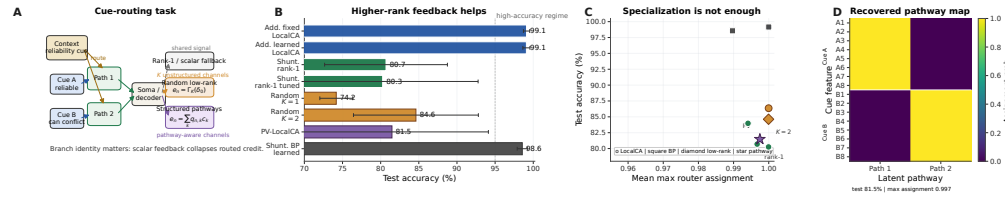


Figure S12: **Structured feedback when rank-1 broadcast fails.** (A) Cue-routing task schematic. Context determines which noisy cue is reliable on each trial, so different dendritic pathways should receive different teaching signals. (B) Five-seed comparison of rank-1, random low-rank, and pathway-structured feedback. Moving beyond rank-1 helps, while the best structured repair remains open. (C) Test accuracy vs. router specialization across learned-router conditions. High specialization alone does not guarantee success. (D) Recovered pathway assignments for a representative learned-router run. Cue-A and cue-B features segregate into separate latent pathways even when feedback quality, rather than router discreteness, is the limiting factor.

## Random Low-Rank Broadcast Channels

## Frozen Activation-Choice Audit

## B Implementation Details

### Biological Plausibility Assumptions

The theorems and rules above rest on a small set of explicit assumptions. We state them as a checklist so that readers can see exactly what is assumed and where each assumption is used:

**A1 (Steady-state voltage).** We use the steady-state solution of the passive cable equation (Eq. 1).

Temporal dynamics are not modeled; all gradients, path gains, and fidelity measurements refer to the conductance-stage voltage at steady state.

**A2 (Tree topology).** Each dendritic cell is a rooted tree with the soma as the root. There are no recurrent connections within the dendrite, which makes the path from any compartment to the soma unique and is what allows the closed-form path gain  $\alpha_n$  in Theorem 1.

**A3 (Nonnegative conductances).** Synaptic and dendritic conductance parameters are pushed through a positive transform (`softplus`), so  $g_j^{\text{syn}}, g_j^{\text{den}} \geq 0$  at every optimisation step. This is a biophysical constraint and is also what keeps the shunting denominator bounded away from zero.

| Setting                                                           | Broadcast                         | Test accuracy                       |
|-------------------------------------------------------------------|-----------------------------------|-------------------------------------|
| <i>Noise resilience, shunting, rank bridge</i>                    |                                   |                                     |
| per-soma                                                          | rank-1 per-soma                   | $0.662 \pm 0.016$                   |
| path propagation                                                  | recursive attenuation proxy       | $0.762 \pm 0.047$                   |
| low-rank                                                          | $K=1$                             | $0.453 \pm 0.126$                   |
| low-rank                                                          | $K=2$                             | $0.630 \pm 0.042$                   |
| low-rank                                                          | $K=4$                             | $0.764 \pm 0.025$                   |
| low-rank                                                          | $K=8$                             | $0.795 \pm 0.019$                   |
| path transport                                                    | exact conductance-stage transport | <b><math>0.847 \pm 0.018</math></b> |
| <i>Cue routing, learned shunting router, rank/structure sweep</i> |                                   |                                     |
| per-soma                                                          | tuned rank-1                      | $0.803 \pm 0.125$                   |
| low-rank                                                          | $K=1$                             | $0.742 \pm 0.022$                   |
| low-rank                                                          | $K=2$                             | <b><math>0.846 \pm 0.082</math></b> |
| low-rank                                                          | $K=4$                             | $0.748 \pm 0.085$                   |
| pathway-vector                                                    | tuned structured rank-2           | $0.815 \pm 0.126$                   |

Table S10: **Rank bridge and rank-structure comparison.** On shunting noise resilience, both the current path-propagation proxy and higher-rank random low-rank broadcast close a substantial fraction of the gap between rank-1 per-soma and exact path transport, with low-rank continuing to improve up to  $K=8$ . On cue routing, the learned-router sweep shows that moving beyond rank-1 helps, but the best current unstructured low-rank setting ( $K=2$ ) is still slightly better than the tuned structured pathway-vector variant. We therefore interpret routed credit assignment as an open higher-rank problem rather than as a solved pathway-vector rescue.

525 **A4 (Nonnegative presynaptic drive).** The first synaptic drive  $x$  is nonnegative, implemented either  
526 by nonnegative inputs or an explicit ReLU transfer stage. Together with A3 this enforces  
527 the positive-input condition used for all headline sweeps.

528 **A5 (Pre-reactivation upward transfer).** Theorem 1 is stated for the pre-reactivation compartment  
529 voltage. A post-voltage reactivation  $f$  (e.g. `param_tanh`) is handled by multiplying the  
530 path gain by  $f'(V_n)$  along the recursion. The compartment error remains the sole non-local  
531 quantity; only the effective path gain acquires local activation-derivative factors.

532 **A6 (Single perisomatic broadcast).** Every synapse receives one non-local scalar or low-rank vector  
533 signal delivered from the soma; all other ingredients of the update rule (eligibility traces,  
534 driving force, input resistance, covariance modulators) are computable from quantities  
535 available at the synapse or compartment.

536 Each synapse therefore has access to: presynaptic activity  $x_j$  (local), compartment voltage  $V_n$  (local  
537 membrane potential), reversal potential  $E_j$  (fixed by receptor type), and total input resistance  $R_n^{\text{tot}} =$   
538  $1/g_n^{\text{tot}}$  (computable from local conductances). The 4F modulator  $\rho_n$  requires online estimation of  
539 voltage covariance (biologically plausible via slow calcium signals); the 5F factor  $\phi_n$  requires a  
540 linear regression proxy (implementable via eligibility traces). The only non-local quantity is the  
541 broadcast error  $e_n$ , which requires a top-down or neuromodulatory signal from the soma to dendritic  
542 compartments. We assume per-neuron resolution (one error per output neuron) for the per-soma  
543 broadcast mode.

#### 544 Dendritic Structure in Code

545 We use the morphology notation  $[b_1, b_2, \dots, b_D]$  for rooted dendritic trees, where each factor gives  
546 the fan-in from one dendritic stage to the next on the path to the soma. For example,  $[3, 3]$  means that  
547 each soma aggregates 3 proximal branches and each proximal branch aggregates 3 distal branches,  
548 for 9 distal leaves per soma;  $[3, 3, 3]$  adds a third dendritic stage with 27 leaves per soma. In code,  
549 each excitatory unit is instantiated as a `DendriNet`, which is a stack of `DendriticBranchLayers`.  
550 Each branch layer contains an excitatory synapse bank, an optional inhibitory synapse bank, a  
551 branch-to-parent aggregation map, and an optional post-voltage reactivation. In the main feedfor-  
552 ward models studied here, `somatic_synapses=false`, so synaptic inputs terminate on dendritic  
553 branches rather than directly on the soma. The headline runs use `input_mode=1`: the transfer layer  
554 creates nonnegative excitatory and inhibitory streams from the same external input, and nonzero

| Dataset / strategy          | Core     | $N_I$ | Identity (none)    | Frozen param_tanh |
|-----------------------------|----------|-------|--------------------|-------------------|
| MNIST / LocalCA             | additive | 0     | $0.780 \pm 0.014$  | $0.709 \pm 0.037$ |
| MNIST / LocalCA             | additive | 10    | $0.907 \pm 0.003$  | $0.881 \pm 0.005$ |
| MNIST / LocalCA             | shunting | 0     | $0.841 \pm 0.007$  | $0.778 \pm 0.020$ |
| MNIST / LocalCA             | shunting | 10    | $0.905 \pm 0.009$  | $0.840 \pm 0.016$ |
| Noise resilience / LocalCA  | additive | 0     | $0.590 \pm 0.162$  | $0.109 \pm 0.006$ |
| Noise resilience / LocalCA  | additive | 10    | $0.859 \pm 0.001$  | $0.772 \pm 0.009$ |
| Noise resilience / LocalCA  | shunting | 0     | $0.411 \pm 0.028$  | $0.331 \pm 0.020$ |
| Noise resilience / LocalCA  | shunting | 10    | $0.769 \pm 0.004$  | $0.701 \pm 0.006$ |
| MNIST / backprop            | additive | 0     | $0.925 \pm 0.001$  | $0.942 \pm 0.001$ |
| MNIST / backprop            | additive | 10    | $0.921 \pm 0.001$  | $0.950 \pm 0.001$ |
| MNIST / backprop            | shunting | 0     | $0.963 \pm 0.0004$ | $0.950 \pm 0.001$ |
| MNIST / backprop            | shunting | 10    | $0.967 \pm 0.001$  | $0.946 \pm 0.001$ |
| Noise resilience / backprop | additive | 0     | $0.896 \pm 0.003$  | $0.106 \pm 0.007$ |
| Noise resilience / backprop | additive | 10    | $0.893 \pm 0.002$  | $0.927 \pm 0.002$ |
| Noise resilience / backprop | shunting | 0     | $0.898 \pm 0.001$  | $0.869 \pm 0.002$ |
| Noise resilience / backprop | shunting | 10    | $0.926 \pm 0.001$  | $0.912 \pm 0.001$ |

Table S11: **Frozen activation-shape audit on matched dendritic architectures (5 seeds per condition).** Identity (none) and a frozen bounded post-voltage reactivation param\_tanh lead to markedly different operating regimes under nonnegative first-layer synaptic drive. In this audit, identity is the safer LocalCA choice on both MNIST and the noise-resilience task, while the backprop effect is strongly regime-dependent: frozen param\_tanh helps additive most clearly in the inhibited noise-resilience setting ( $N_I=10$ ) but collapses it at  $N_I=0$ . These runs match additive reactivation initialization to the shunting setting and hold the activation parameters fixed, isolating activation shape from both additive/shunting initialization differences and activation learning itself.

| Dataset          | Core     | $N_I$ | Identity (none)        | Frozen param_tanh      |
|------------------|----------|-------|------------------------|------------------------|
| MNIST            | additive | 0     | CV 0.280, cosine 0.220 | CV 1.122, cosine 0.074 |
| MNIST            | additive | 10    | CV 0.310, cosine 0.212 | CV 1.079, cosine 0.094 |
| MNIST            | shunting | 0     | CV 0.393, cosine 0.170 | CV 0.770, cosine 0.114 |
| MNIST            | shunting | 10    | CV 0.385, cosine 0.281 | CV 1.177, cosine 0.106 |
| Noise resilience | additive | 0     | CV 0.124, cosine 0.079 | CV 0.114, cosine 0.052 |
| Noise resilience | additive | 10    | CV 0.414, cosine 0.206 | CV 0.840, cosine 0.094 |
| Noise resilience | shunting | 0     | CV 0.255, cosine 0.324 | CV 0.344, cosine 0.341 |
| Noise resilience | shunting | 10    | CV 0.762, cosine 0.140 | CV 2.497, cosine 0.079 |

Table S12: **Frozen activation-shape theory audit on matched LocalCA runs (5 seeds per condition).** Each entry reports conductance-stage path-gain dispersion (CV) and direct per-soma compartment-error fidelity (cosine between the per-soma broadcast and the exact pre-activation compartment error). The positive-input theory audit largely agrees with the empirical one: freezing param\_tanh usually broadens path-gain dispersion and weakens per-soma fidelity, especially in the inhibited settings where the main mechanistic story lives. The one mild exception is low-inhibition shunting on noise resilience, where frozen param\_tanh leaves fidelity roughly unchanged while still hurting final accuracy, suggesting that activation choice affects optimization and representation as well as direct credit fidelity.

ie\_synapses\_per\_branch\_per\_layer attaches learned inhibitory conductance banks directly to excitatory dendritic branches. These models therefore have input-driven branch inhibition but no separate inhibitory neuron population. By contrast, input\_mode=0 instantiates explicit feedforward inhibitory cells when inhibitory\_layer\_sizes is nonempty; these cells can transform excitatory activity before projecting inhibitory conductance back onto excitatory dendrites. In that setting the inhibitory cells are also dendritic modules, so their E-to-I and I-to-I synaptic conductances are trained by the same local rule family rather than by a separate global inhibitory optimizer. We treat that explicit-I setting as an exploratory mechanism probe rather than a headline baseline, because the main claims were measured in the direct-I-stream regime. The additive and shunting cores



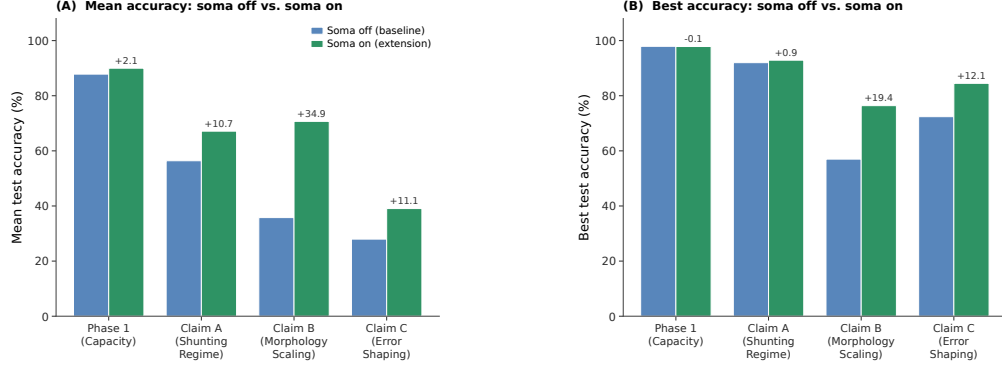


Figure S13: **Soma-on extension for LocalCA.** Enabling direct somatic input (`somatic_synapses=true`) improves aggregate mean test accuracy across all four manuscript LocalCA families. Each family is summarized over the completed configurations in matched soma-off and soma-on sweeps. The largest gain appears in morphology scaling, where the family mean rises from 35.8% to 70.7% and the best run rises from 57.0% to 76.4%. Gains in the shunting-regime and error-shaping families are smaller but still substantial. Delta labels show soma-on minus soma-off. Because the main mechanistic analysis was developed in the dendrite-only feedforward regime, we report soma-on as a supportive extension rather than replacing the main operating regime.

| Family             | Soma-off | Soma-on | $\Delta$ mean | $\Delta$ best |
|--------------------|----------|---------|---------------|---------------|
| Phase-1 capacity   | 0.8784   | 0.8999  | +0.0215       | -0.0007       |
| Shunting regime    | 0.5641   | 0.6715  | +0.1074       | +0.0090       |
| Morphology scaling | 0.3578   | 0.7070  | +0.3492       | +0.1940       |
| Error shaping      | 0.2798   | 0.3906  | +0.1109       | +0.1210       |

Table S13: **Representative soma-on deltas across the four LocalCA families.** Values compare matched families with and without `somatic_synapses` and report both the change in family-mean test accuracy and the change in best-run test accuracy. Cue-routing soma and CIFAR-10 depth-4 soma are reported separately in Fig. S14, Fig. S9, and Table S7 because those extensions are summarized on different task families.

are architecture-matched at the tree level; they differ in the branch-level voltage integration rule rather than in whether a dendritic tree exists. In all reported dendritic runs, synaptic and dendritic conductance parameters are pushed through positive transforms (`softplus` in the main paper runs), so the conductances themselves remain nonnegative at every optimization step; unconstrained decoder weights are separate readout parameters and are not interpreted as conductances.

Matched soma-on LocalCA sweeps suggest that direct somatic input is a meaningful extension rather than a minor implementation detail. Relative to matched non-somatic sweeps, enabling `somatic_synapses=true` raises the mean test accuracy in all four manuscript families: phase-1 capacity from 0.8784 to 0.8999, shunting-regime sweeps from 0.5641 to 0.6715, morphology-scaling sweeps from 0.3578 to 0.7070, and error-shaping sweeps from 0.2798 to 0.3906. These numbers are family-level aggregates over the completed configurations in each matched soma-off and soma-on sweep, rather than a single 5-seed cell. Because the main mechanistic analysis in this paper was developed in the dendrite-only feedforward regime, we keep the main text anchored to `somatic_synapses=false` and treat soma-on as a supportive extension rather than silently replacing the original operating regime. Figure S13 summarizes the mean and best-accuracy comparisons across the four families.

The soma-on benefit extends beyond the capacity/scaling families to the cue-routing benchmark that stress-tests rank-1 feedback. With dendrite-only pathways, the best learned-routing shunting LocalCA configuration was unstable at  $0.81 \pm 0.13$ ; with `somatic_synapses=true`, fixed-router shunting LocalCA reaches  $1.00 \pm 0.0$  test accuracy across 5 seeds, and learned-router additive LocalCA reaches  $0.99 \pm 0.003$  (Fig. S14). The soma channel provides a high-bandwidth direct input that bypasses the rank-1 bottleneck induced by per-soma broadcast.

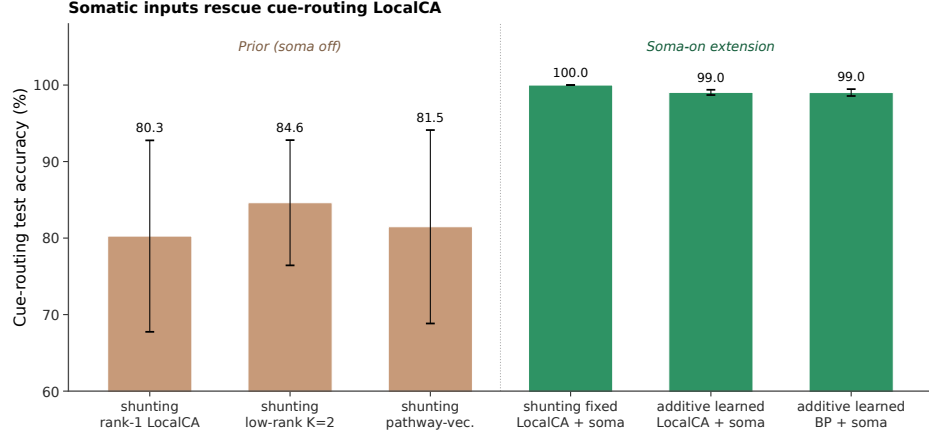


Figure S14: **Soma-on rescues cue-routing LocalCA.** The left bars show the previously reported soma-off LocalCA results on the cue-integration benchmark. The right bars show the soma-on extension: fixed-router shunting LocalCA reaches 1.00 test accuracy, and learned-router additive LocalCA with soma reaches 0.99, matching the additive backprop ceiling with soma. The soma channel supplies a direct high-bandwidth input that bypasses the rank-1 bottleneck identified in the main text. Learned-router shunting with soma is omitted because this benchmark uses signed cues, which conflict with the shunting positive-input check in the learned-router pipeline; fixed-router shunting is therefore the clean shunting data point.

## 586 Post-Voltage Reactivation

587 The conductance stage produces a pre-reactivation compartment voltage  $V_n$ , after which each branch  
 588 layer may apply a monotone scalar nonlinearity. The implementation supports identity (none), fixed  
 589  $\tanh/\text{sigmoid}/\text{relu}$ , and several parametric families including `param_tanh`, `param_tanh_only_m`,  
 590 and piecewise linear-saturating variants. The main manuscript sweeps use `param_tanh`, a learnable  
 591 bounded firing-rate transform of the form  $(\tanh(m(V - b)) + 1)/2$ , together with an explicitly  
 592 configured nonnegative transfer stage so that the first synaptic drive remains nonnegative even  
 593 after input corruption or structured task noise. Most headline feedforward sweeps use a linear  
 594 decoder, while the CIFAR extension uses a nonlinear decoder and therefore maps output error back to  
 595 decoder-input/soma space through the decoder Jacobian before applying LocalCA. Unless explicitly  
 596 overridden, additive and shunting cores can start from different reactivation initializations; activation-  
 597 fairness ablations therefore match additive initialization to the shunting setting when comparing  
 598 alternative reactivation choices directly. The appendix activation audit additionally fixes `param_tanh`  
 599 after initialization to isolate activation shape from activation plasticity.

600 LocalCA uses occupancy-quantile initialization together with a calibration pass that floors tiny  
 601 quantile widths, clamps extreme fitted slopes, and reverts invalid per-layer fits rather than allowing  
 602 pathological intermediate values to propagate. In the non-somatic feedforward publication families,  
 603 this calibration leaves the results essentially unchanged relative to the same occupancy-quantile setup  
 604 without the additional validity checks: phase-1 capacity changes from 0.8786 to 0.8784 mean test  
 605 accuracy, shunting-regime sweeps from 0.5617 to 0.5641, morphology-scaling sweeps from 0.3600  
 606 to 0.3578, and error-shaping sweeps from 0.2860 to 0.2798. The calibration therefore improves  
 607 numerical safety without changing the qualitative LocalCA story.

608 We also compared two post-initialization policies for the reactivation parameters under quantile  
 609 initialization: *learned local* ( $b, m$ ), where LocalCA updates the gate parameters through the local  
 610 rule, and *quantile-maintained* ( $b, m$ ), where  $(b, m)$  are periodically refreshed from the current voltage  
 611 distribution instead of being updated by the local rule. This comparison matters because quantile-  
 612 maintained gates substantially simplify nonlinear reactivation learning: the rest of the network still  
 613 uses LocalCA, but the gate parameters themselves are set by a local distributional rule rather than  
 614 a learned plasticity rule. On MNIST classification sweeps, learned local ( $b, m$ ) remains clearly  
 615 better (additive 0.9247 vs. 0.9061 mean test accuracy; shunting 0.9104 vs. 0.8766). On CIFAR-10,  
 616 the picture is mixed in a useful way: additive is essentially tied on mean (0.4223 learned-local vs.  
 617 0.4215 quantile-maintained) while shunting improves under quantile-maintained gates (0.4028 to

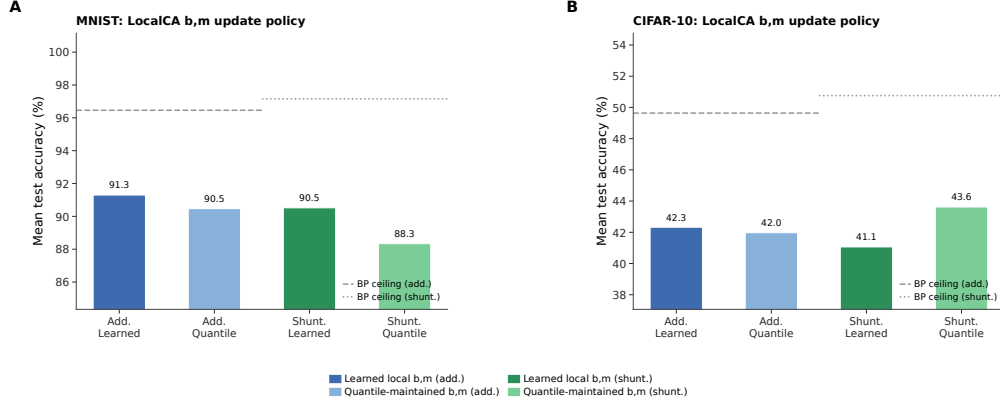


Figure S15:  $(b, m)$  update policy under quantile initialization. Each cluster of bars compares the mean test accuracy of learned local  $(b, m)$  and quantile-maintained  $(b, m)$  under LocalCA for additive and shunting cores, averaged across a matched morphology/init grid  $([2, 2]$  and  $[3, 3, 3]$ ; analytical and quantile init variants). Dashed and dotted horizontal lines mark the matched standard backpropagation ceiling for additive and shunting respectively. **Left (MNIST):** Learned local  $(b, m)$  is consistently better; quantile-maintained lags by 1.9 pp (additive) and 3.4 pp (shunting) on mean accuracy. **Right (CIFAR-10):** The gap reverses for shunting: quantile-maintained reaches 43.6% mean vs. 40.3% for learned local, while additive is essentially tied. The pattern suggests that quantile-maintained  $(b, m)$  prevents over-fitting the reactivation gate on harder inputs, making it a useful alternative in harder visual regimes.

| Quantity         | Symbol                           | Convention              |
|------------------|----------------------------------|-------------------------|
| Voltage          | $V$                              | Normalized to $[-1, 1]$ |
| Conductances     | $g^{\text{syn}}, g^{\text{den}}$ | Nonneg. via softplus    |
| Leak conductance | $g^{\text{leak}}$                | Set to 1                |
| Input resistance | $R^{\text{tot}}$                 | $\leq 1$                |

Table S14: Units and normalization.

0.4356 mean test accuracy). These bars aggregate over a matched morphology/init grid rather than a dedicated 5-seed rerun, so we treat them as a supportive policy comparison rather than a headline result. Accordingly, the main manuscript continues to use occupancy-quantile initialization followed by LocalCA updates of  $(b, m)$  as the default local-learning setting, while quantile-maintained  $(b, m)$  is reported as a supplementary alternative. Figure S15 shows the full MNIST and CIFAR-10 comparison.

## Component Ablation

To isolate the contribution of each ingredient in our LocalCA recipe, we ran a per-component ablation on MNIST starting from the manuscript default and removing one component at a time. The default combines quantile init, learned local  $(b, m)$ , `somatic_synapses=false`, and `param_tanh` reactivation. We also include `somatic_synapses=true` as a positive extension and matched standard backpropagation as the upper reference. Figure S16 reports test accuracy (5 seeds per cell) for both shunting and additive dendritic cores on clean MNIST with morphology  $[3, 3]$ .

## Units and Parameterization

### Decoder Update Modes

$W_{\text{dec}}$  maps  $V_L \rightarrow \hat{y}$ . Modes: **backprop** ( $\nabla_W L$  via autograd), **local** ( $\Delta W = \eta \langle \delta_0 V_L^T \rangle_B$ ), **frozen** ( $\Delta W = 0$ ).

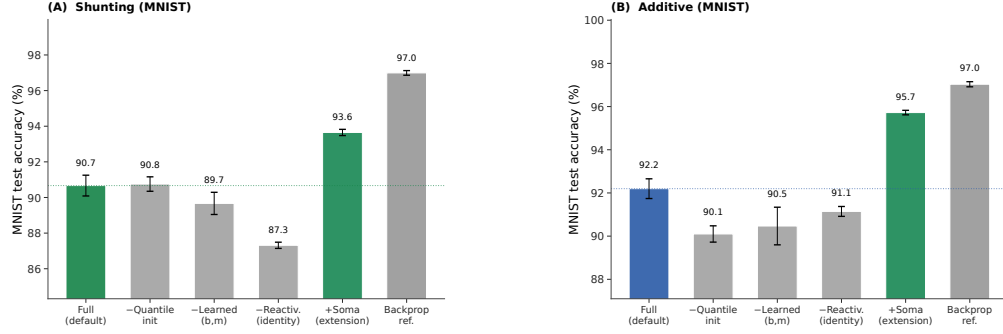


Figure S16: **Per-component ablation for LocalCA on MNIST.** Bars show mean test accuracy ( $\pm$  s.d. across 5 seeds); coloured bars highlight the manuscript default (blue/green) and the soma-on extension (green). **(A) Shunting:** The full default reaches 90.7%; removing quantile init leaves accuracy unchanged at 90.8% (quantile init is a mild calibration nudge on this clean task), removing learned ( $b, m$ ) drops by about 1 pp, and removing reactivation entirely drops accuracy to 87.3%; the reactivation transform is the largest single LocalCA ingredient. **(B) Additive:** Quantile init contributes +2.1 pp (90.1  $\rightarrow$  92.2%), learned ( $b, m$ ) contributes +1.7 pp, and reactivation contributes +1.1 pp; each ingredient helps on the additive core. **Soma-on** is the largest positive change in both cores (+3 to +3.5 pp over full config), while matched standard backpropagation reaches 97.0% in both cores and sets the upper reference. The fixed-relu reactivation condition did not satisfy the same initialization checks and is omitted.

| Category                    | Implemented modes                                                                                | Role in this paper                                                                                                                                                                                  |
|-----------------------------|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Core models                 | point MLP; dendritic MLP; dendritic additive; dendritic shunting; pathway-routed dendritic cores | Main comparisons use dendritic additive vs. dendritic shunting; routed cores are used for cue integration.                                                                                          |
| Training strategies         | standard backprop; LocalCA; DFA; FA; transported-broadcast oracle                                | Main text uses standard and LocalCA; DFA/FA remain appendix baselines; path transport is an oracle upper bound.                                                                                     |
| Broadcast modes             | scalar; per-soma; local mismatch; low-rank; path transport; pathway vector                       | Main text centers on per-soma, low-rank, and transported-error controls; scalar and local-mismatch are supporting controls, while pathway vector remains an appendix structured-feedback extension. |
| Recent dendritic extensions | path propagation; router-derived branch roles; pathway-vector feedback; inhibitory homeostasis   | These were added to test morphology-aware credit transport and structured branch-specific plasticity beyond the original 3F/4F/5F family.                                                           |

Table S15: Implemented model families and LocalCA operating modes. Only a subset is used for the main claims; the remaining modes serve as controls, ablations, or structured extensions.

## 635 Model and Mode Inventory

## 636 Representative Manuscript Architectures

## 637 Hyperparameters

638 Table S17 collects the core training hyperparameters for each headline experiment. Learning rates  
639 are applied through parameter groups (top- $k$ , dendritic block-linear, reactivation, decoder) with the  
640 values given in the representative config files; where a single LR is reported, every group uses that  
641 value.

| Setting                                                    | Encoder  | E layers           | Tree                                         | Synapses / branch                      | Seeds                | Notes                                                                                                             |
|------------------------------------------------------------|----------|--------------------|----------------------------------------------|----------------------------------------|----------------------|-------------------------------------------------------------------------------------------------------------------|
| Gradient fidelity / path-gain / noise-resilience mechanism | id.+ReLU | [128, 128]         | [3, 3]                                       | $N_E=40, N_I \in \{0, 5, 10, 20, 40\}$ | 5 fidelity, 5 oracle | Main mechanism sweeps in Figs. 2 and 3; reactivation <code>param_tanh</code> .                                    |
| MNIST / context-gating local competence                    | id.+ReLU | [128]              | [3, 3]                                       | $N_E=40, N_I=20$                       | 5                    | Best local 5F per-soma runs in Table S4; reactivation <code>param_tanh</code> .                                   |
| Fashion-MNIST competence                                   | id.+ReLU | [128]              | [3, 3]                                       | $N_E=40, N_I=20$                       | 5                    | Dedicated five-seed standard-vs-local competence sweep; reactivation <code>param_tanh</code> .                    |
| Cue-routing PV-LocalCA                                     | router   | [64]               | [2]                                          | $N_E=10, N_I=4$                        | 5                    | Learned router with two pathway groups and rank-2 structured feedback; reactivation <code>param_tanh</code> .     |
| CIFAR-10 compact direct-I-stream mechanism extension       | id.+ReLU | [20] E, no I cells | [3, 3, 3, 3]                                 | $N_E=25, N_I=25$ direct I-to-E         | 5                    | Harder-data extension with input-driven inhibitory conductance, decoder [32, 16], and decoder-aware soma mapping. |
| Morphology $\times$ inhibition appendix map                | id.+ReLU | [128, 128]         | [2, 2], [3, 3], [4, 4], [2, 2, 2], [3, 3, 3] | $N_E=40, N_I \in \{0, 5, 10, 20, 40\}$ | 3                    | Supportive 3-seed sweep; reactivation <code>param_tanh</code> .                                                   |

Table S16: **Representative architectures used in the manuscript.** All dendritic rows use explicit branch-level excitatory and inhibitory synapse banks with `somatic_synapses=false`, nonnegative conductances, and a nonnegative first-layer drive enforced by an explicit ReLU transfer stage in the main runs. The tree column gives dendritic morphology per excitatory neuron, and the synapse counts report excitatory and inhibitory synapses per branch.

## Effective Broadcast Dimensionality in Headline Architectures

### LocalCA Extension Compatibility

### Seed Protocol and Evidence Map

The seed depth is not uniform across every sweep, and we therefore separate main evidence from supporting diagnostics. Table S20 pins each figure or table in this paper to its seed count and evidence tier.

## Compute Resources

All experiments were run on an internal GPU cluster. Most MNIST, Fashion-MNIST, and synthetic-task runs used a single NVIDIA A100 or H100 GPU and finished in minutes to tens of minutes per seed. The flattened CIFAR-10 runs required longer single-GPU jobs but still did not use distributed or multi-GPU training. Reported headline metrics are aggregated over 5 seeds unless noted otherwise. The remaining 3-seed results are confined to selected supportive sweeps, and the broader exploratory screens used to tune and diagnose the method required additional compute beyond the final reported runs.

## Code, Data, and Asset Availability

MNIST, Fashion-MNIST, CIFAR-10, and MICrONS are public datasets or public benchmark assets cited in the paper; we use them under their standard published access conditions and cite their original sources in the bibliography. The present anonymous submission does not include a public code URL in order to preserve double-blind review, but the paper specifies the model family, training modes, key hyperparameters, and representative configurations, and we plan to release code and configuration files after review. The paper does not release a new dataset or a stand-alone pretrained model asset.

| Setting                                                      | Opt. | LR                                              | Batch | Epochs    | W-dec.   | Notes                                                                |
|--------------------------------------------------------------|------|-------------------------------------------------|-------|-----------|----------|----------------------------------------------------------------------|
| MNIST / Fashion-MNIST / context gating (5F per-soma LocalCA) | Adam | $10^{-3} / 5 \cdot 10^{-4}$<br>(block / react.) | 256   | 100       | 0        | fixed-epoch, no early stopping; <code>param_tanh</code> reactivation |
| MNIST / Fashion-MNIST / context gating (BP ceiling)          | Adam | $10^{-3} / 5 \cdot 10^{-4}$                     | 256   | 100       | 0        | matched to LocalCA setup                                             |
| Gradient fidelity / path-gain / noise resilience             | Adam | $10^{-3}$                                       | 256   | 50        | 0        | 5 seeds, hooks enabled for diagnostic capture                        |
| Morphology $\times$ inhibition regime map (exploratory)      | Adam | $10^{-3}$                                       | 256   | 50        | 0        | 3 seeds                                                              |
| Cue routing (soma-off & soma-on)                             | Adam | $10^{-3}$                                       | 256   | 90        | 0        | early stopping with patience 30                                      |
| CIFAR-10 compact direct-I-stream depth-4 LocalCA             | Adam | $10^{-3} / 10^{-4}$<br>(block / react.)         | 256   | 400       | 0.01     | grad-clip 5.0, early stop patience 50, nonlinear decoder             |
| CIFAR-10 compact direct-I-stream depth-4 BP                  | Adam | $10^{-3}$                                       | 256   | 400       | 0.01     | matched ceiling                                                      |
| Weight-distribution vs. depth                                | Adam | $10^{-3}$                                       | 256   | 100       | 0        | <code>weight_analysis</code> hook captures final-epoch statistics    |
| Component ablation (MNIST)                                   | Adam | $10^{-3}$                                       | 256   | 100       | 0        | 5 seeds, single-layer dendrinet [3, 3]                               |
| $(b, m)$ policy comparison (MNIST / CIFAR-10)                | Adam | $10^{-3} / 10^{-4}$                             | 256   | 100 / 400 | 0 / 0.01 | quantile refresh every 5 epochs; EMA $\alpha \in \{0.25, 0.5, 1.0\}$ |

Table S17: **Training hyperparameters by experiment.** All runs use Adam with default  $\beta_1=0.9$ ,  $\beta_2=0.999$ ,  $\varepsilon=10^{-8}$ . Reactivation and block-linear parameter groups use a slower LR to keep conductance-level dynamics stable. Full configs are provided under `drafts/dendritic-local-learning/configs/`.

| Setting                                          | Hidden width(s) | Output dim           | Effective default per-soma field                                                                              |
|--------------------------------------------------|-----------------|----------------------|---------------------------------------------------------------------------------------------------------------|
| Gradient fidelity / path-gain / noise resilience | 128, 128        | 10                   | Scalar / rank-1 at both dendritic layers                                                                      |
| MNIST / Fashion-MNIST competence                 | 128             | 10                   | Scalar / rank-1 at the dendritic hidden layer                                                                 |
| Context gating competence                        | 128             | 2                    | Scalar / rank-1 at the dendritic hidden layer                                                                 |
| Cue routing                                      | 64              | 2                    | Scalar / rank-1 by default; higher-rank tests use explicit low-rank or pathway-vector feedback                |
| Compact CIFAR-10 harder-data extension           | 20              | decoder-input dim 20 | Vector-valued per-soma after decoder-aware mapping; higher-rank tests use explicit low-rank or path transport |

Table S18: **Effective dimensionality of the default “per-soma” broadcast in the headline architectures.** In code, per-soma uses the output error vector only when layer width matches decoder output dimension; otherwise it falls back to a shared scalar. The main feedforward settings in this paper therefore test a genuinely rank-1 broadcast regime by default, which is why the higher-rank low-rank, path-transport, and pathway-vector controls are important.

## 663 Algorithm

### Algorithm 1 Local Credit Assignment

- 1: **Input:** Model, batch  $(x, y)$ , config  $\mathcal{C}$
- 2: Forward pass; loss  $L$ , output error  $\delta^y$
- 3: Somatic/core error  $\delta_0 = \text{decoder-input Jacobian product } J_{\text{dec}}(h_{\text{core}})^\top \delta^y$  (or  $W_{\text{dec}}^\top \delta^y$  for a linear decoder)
- 4: **for** each layer  $n$  (reverse) **do**
- 5:    $e_n = \text{broadcast}(\delta_0, \mathcal{C})$ ; optionally gate by pathway roles
- 6:   Compute  $\rho_n, \phi_n$  (EMA estimators, if enabled)
- 7:   Apply the configured local update (Eq. 10 or 11)
- 8: **end for**
- 9: Clip gradients; optimizer step

| Extension              | Meaning                                                              | Compatible with | Typical use here                                                                               |
|------------------------|----------------------------------------------------------------------|-----------------|------------------------------------------------------------------------------------------------|
| Per-soma broadcast     | Somatic error vector when widths match, otherwise scalar fallback    | 3F, 4F, 5F      | Default main-text LocalCA setting; in the headline feedforward runs this is effectively rank-1 |
| Low-rank broadcast     | Small number of broadcast channels instead of a scalar or full field | 3F, 4F, 5F      | Intermediate control when rank-1 is too restrictive                                            |
| Path transport         | Recursive tree-transported error using conductance-stage path gains  | 3F, 4F, 5F      | Oracle upper-bound broadcast                                                                   |
| Path propagation       | Approximate path-gain attenuation applied to broadcast               | 3F, 4F, 5F      | Morphology-aware modifier for deeper trees                                                     |
| Pathway vector         | Structured branch/pathway-specific broadcast field                   | 3F, 4F, 5F      | Used mainly in PV-LocalCA on routed tasks                                                      |
| Branch-role rules      | Scales updates by router- or pathway-derived branch specialization   | 3F, 4F, 5F      | Used in the structured pathway extension                                                       |
| Inhibitory homeostasis | Auxiliary local inhibitory balancing term                            | 3F, 4F, 5F      | Optional stabilizer in routed or shunting settings                                             |

Table S19: Compatibility of the newer LocalCA extensions with the base 3F/4F/5F family. These extensions mostly modify the error field or branch scaling rather than defining a new “6F” or “7F” rule.

## C Theoretical Details

### Variant Taxonomy

#### Random-Broadcast Alignment

**Proposition 5** (Supportive random-broadcast alignment). *Let the broadcast matrix  $B_n$  have i.i.d. zero-mean entries with  $\mathbb{E}[B_n^\top B_n] = \alpha I$ . Then the expected cosine between the resulting local gradient and the exact gradient is positive,  $\mathbb{E}[\cos \angle(g^{\text{local}}, g^{\text{exact}})] \geq c_n > 0$ , by an argument analogous to feedback alignment [9]. The constant  $c_n$  depends on how strongly local eligibility factors correlate with the conductance-stage path gain in Eq. (4); shunting improves this correlation by narrowing the spread of intermediate sensitivities.*

We treat this argument as supportive rather than central. The main theoretical object in the paper is still the exact pre-reactivation compartment error  $\partial L / \partial V_n$ ; under identity upward transfer this reduces to  $\alpha_n \delta_0$ , while with post-voltage activation the corresponding effective path gain additionally includes local  $f'_n(V_n)$  factors. The main empirical question is how faithfully different broadcast fields approximate that quantity.

#### Biological Analogs

## D Morphology-Aware Extensions

**Path-integrated propagation.** Modulate broadcast error by  $\pi_n = \pi_{n-1} \cdot R_{n-1}^{\text{tot}} \cdot \bar{g}_{n-1}^{\text{den}}$ , approximating depth attenuation from Eq. 4.

**Depth modulation.** Per-branch scaling  $\rho_j = \rho_{\text{base}} / (d_j + \alpha)$ , mirroring cable attenuation.

**Dendritic normalization.**  $\Delta g_j^{\text{den}} \leftarrow \Delta g_j^{\text{den}} / (\sum_k g_k^{\text{den}} + \varepsilon)$ , analogous to homeostatic scaling [21].

| Figure / Table             | Sweep                                                | Replication                       | Tier        |
|----------------------------|------------------------------------------------------|-----------------------------------|-------------|
| Fig. 1                     | Model, rule, and broadcast definitions               | N/A                               | Definitions |
| Fig. 2                     | Gradient-fidelity diagnostic                         | 5 seeds                           | Headline    |
| Fig. 3                     | Mechanistic chain + oracle + summary                 | 5 seeds                           | Headline    |
| Fig. 4                     | Competence / regime dose-response                    | 5 seeds                           | Headline    |
| Fig. S8 / Table S6         | CIFAR-10 compact direct-I-stream mechanism extension | 5 seeds                           | Supportive  |
| Fig. S12                   | Cue-routing (soma off)                               | 5 seeds                           | Supportive  |
| Table S4                   | Competence gaps (BP ceiling vs. 5F)                  | 5 seeds                           | Headline    |
| Fig. S13                   | Soma-on extension across manuscript families         | Family aggregates (144–420 cfigs) | Supportive  |
| Table S13                  | Soma-on delta summary across manuscript families     | Family aggregates                 | Supportive  |
| Fig. S15                   | $(b, m)$ update-policy comparison                    | Matched 4-cell grid / bar         | Supportive  |
| Fig. S16                   | Component ablation (MNIST)                           | 5 seeds                           | Supportive  |
| Fig. S14                   | Cue-routing soma-on (stable subset)                  | 5 seeds                           | Supportive  |
| Table S9                   | Fitted direct-I inhibitory path-gain diagnostic      | 5 runs                            | Supportive  |
| Fig. S17                   | Weight-distribution sensitivity                      | 5 seeds                           | Supportive  |
| Fig. S18                   | Weight-distribution vs. depth                        | 5 seeds                           | Supportive  |
| Fig. S19                   | Weight-distribution soma-off vs. soma-on             | 5 seeds                           | Supportive  |
| Fig. S9 / Table S7         | CIFAR-10 depth-4 soma extension                      | 3 seeds                           | Supportive  |
| Fig. S7                    | FA/DFA baseline comparison                           | 5 seeds                           | Supportive  |
| Fig. S6                    | Broadcast-bandwidth / quantization                   | 5 seeds                           | Supportive  |
| Fig. S10                   | Additive + normalization control                     | 5 seeds                           | Supportive  |
| Fig. S5 / S4               | Stress tests; verification                           | 5 seeds                           | Supportive  |
| Fig. S3                    | 5F sensitivity                                       | 3 seeds                           | Supportive  |
| Table S10                  | Rank-bridge                                          | 5 seeds                           | Supportive  |
| Fig. 5                     | Morphology $\times$ inhibition map                   | 3 seeds                           | Supportive  |
| Fig. S11 / Table S8        | One-layer direct-I vs. explicit-I input-mode probe   | 3 seeds                           | Supportive  |
| Signed-input info-shunting | Signed-input stress test                             | 1–3 seeds                         | Exploratory |

**Table S20: Evidence map.** Each main-paper and appendix figure or table is pinned to its sweep and replication type. Most main claims come from dedicated 5-seed sweeps; the supporting soma-extension and  $(b, m)$  policy appendices instead summarize matched sweep families or matched config grids, and are labeled accordingly. The BP ceilings in Table S4 come from dedicated matched-capacity 5-seed standard-training refreshes.

**Pathway-vector feedback.** For tasks with latent pathway structure, the broadcast becomes a role vector inferred from the router. Local pathway activity gates this vector. Upward block transport then passes it to earlier branches, aligning their feedback with downstream descendants.

**Router-derived branch roles.** Each branch receives a role profile inferred from router assignments and incoming block weights rather than from a hand-coded apical/basal label. Plasticity is then modulated by branch selectivity and a synapse-specific role-alignment factor, emphasizing pathway-consistent updates without imposing a heuristic branch taxonomy.



| Asset           | Use in this paper                                                   | Access / license note                                                                                                                        |
|-----------------|---------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| MNIST           | Digit classification and derived nonnegative synthetic tasks        | Public academic benchmark credited to the original dataset authors; used through standard public access, with no raw-data redistribution.    |
| Fashion-MNIST   | Apparel classification stress test                                  | Public benchmark released with the Fashion-MNIST paper and repository; used through standard public access, with no raw-data redistribution. |
| CIFAR-10        | Flattened harder-data diagnostic                                    | Public research benchmark from the CIFAR dataset collection; used through standard public access, with no raw-data redistribution.           |
| MICrONS         | Biological reference statistics in the weight-distribution appendix | Public connectomics resource cited to the MICrONS Consortium; only aggregate comparison statistics are reported here.                        |
| Synthetic tasks | Context gating, noise resilience, cue integration                   | Generated procedurally from public benchmarks or random seeds described in Appendix I; no new third-party asset is introduced.               |

Table S21: **Existing assets used in the manuscript.** We credit the original sources in the bibliography, use public benchmark assets under their published access conditions, and do not redistribute third-party raw data in the anonymous submission.

| Rule       | Factors                                               | Cost               | Best regime                  |
|------------|-------------------------------------------------------|--------------------|------------------------------|
| 3F         | $x, (E - V), e$                                       | $\mathcal{O}(1)$   | Baseline                     |
| 4F         | $3F + \rho$                                           | $\mathcal{O}(1)$   | Improved conditioning        |
| 5F         | $4F + \phi$                                           | $\mathcal{O}(d_n)$ | Strongest overall            |
| 5F+VH      | $5F + \text{local voltage homeostasis}$               | $\mathcal{O}(d_n)$ | Stabilized LocalCA / warm-up |
| PV-LocalCA | $3F + \text{structured } e_n + \text{role alignment}$ | $\mathcal{O}(r_n)$ | Pathway discovery            |

Table S22: Variant taxonomy. ‘5F+VH’ denotes the same base 5F rule with an additional strictly local voltage-centering auxiliary that can also be scheduled through multi-stage LocalCA warm-up; the codebase also supports this as a shorthand ‘5f\_vh’ alias in configuration files.

## E HSIC Auxiliary Objectives

Following [17], we use kernel-based HSIC objectives:

$$\mathcal{L}^{\text{self}} = B^{-2} \text{tr}(\mathbf{K}_Z \mathbf{H} \mathbf{K}_Z \mathbf{H}), \quad \mathcal{L}^{\text{target}} = -B^{-2} \text{tr}(\mathbf{K}_Z \mathbf{H} \mathbf{K}_Y \mathbf{H}).$$

Moderate weights (0.01–0.1) help on context gating; negligible on MNIST. Online statistics ( $\rho_n, \phi_n$ ) use Welford’s algorithm [20].

## F Online Variant with Eligibility Traces

The continuous-time eligibility trace is

$$\tau_e \dot{e}_j^{\text{syn}} = -e_j^{\text{syn}} + x_j(E_j - V_n)R_n^{\text{tot}},$$

with update

$$\Delta g_j^{\text{syn}} \propto \int e_j^{\text{syn}}(t) m_n(t) dt$$

[18, 19].

## G Weight Statistics and Biological References

As a secondary analysis, we examine whether the conductance-based update naturally induces characteristic weight statistics. Because synaptic updates scale with input resistance  $R_n^{\text{tot}} = 1/g_n^{\text{tot}}$ , the volt-

| Component          | Analog                         | Interpretation                         |
|--------------------|--------------------------------|----------------------------------------|
| $R_n^{\text{tot}}$ | Input resistance               | Sensitivity modulation                 |
| $(E_j - V_n)$      | Synaptic driving force         | Local gradient factor                  |
| Shunting           | Divisive normalization         | $\partial V / \partial g_I \propto -V$ |
| $\rho_n$           | Layer relevance                | Output correlation                     |
| $\phi_n$           | Signal propagation             | Conditional predictability             |
| Pathway vector     | Branch-specific top-down drive | Structured teaching signal             |
| Role alignment     | Pathway-selective plasticity   | Branch specialization                  |

Table S23: Biological analogs.

| Condition        | MNIST                                                      |              | Fashion-MNIST |              |
|------------------|------------------------------------------------------------|--------------|---------------|--------------|
|                  | Exc $\sigma$                                               | Inh $\sigma$ | Exc $\sigma$  | Inh $\sigma$ |
| BP + Shunting    | 0.94±0.01                                                  | 0.87±0.01    | 0.92±0.01     | 0.84±0.01    |
| BP + Additive    | 1.30±0.01                                                  | 1.22±0.01    | 1.16±0.01     | 1.05±0.01    |
| Local + Shunting | 1.22±0.01                                                  | 1.71±0.02    | 1.13±0.06     | 1.36±0.06    |
| Local + Additive | 1.69±0.16                                                  | 1.57±0.07    | 1.43±0.16     | 1.11±0.12    |
| Song et al. 2005 | $\sigma = 0.94$ (rat V1 L5, EPSP amplitude)                |              |               |              |
| MICrONS E→E      | $\sigma = 1.14$ (mouse V1, synapse size, $n=3.9\text{M}$ ) |              |               |              |
| MICrONS I→E      | $\sigma = 0.92$ (mouse V1, synapse size, $n=3.5\text{M}$ ) |              |               |              |

Table S24: **Weight distribution width.** Log-normal  $\sigma$  of excitatory and inhibitory synaptic weights (mean±std across 5 seeds). Biological references are included only as descriptive anchors. With biologically constrained E/I ratios (4:1), shunting self-normalization produces weight heterogeneity in the range spanned by those references, whereas additive integration is broader and less stable.

age equation implies multiplicative weight dynamics, creating a regime analogous to exponentiated-gradient updates. Such multiplicative dynamics suggest log-normal weight distributions in trained networks [38, 39]. We compare the resulting distributions against two biological references: electrophysiological data from Song et al. [38] (rat V1 L5 pyramidal,  $\sigma = 0.94$ ) and the MICrONS mm<sup>3</sup> connectome [40] (mouse V1 synapse sizes).

**Model weight distributions are log-normal.** We fit log-normal distributions to the active (top- $K$  masked) excitatory synaptic weights after training on MNIST and Fashion-MNIST. Table S24 shows the fitted  $\sigma$  (standard deviation of  $\log w$ ) for each condition. Shunting consistently produces narrower distributions than additive integration, and backpropagation produces narrower distributions than local learning. These two factors combine approximately additively: BP+Shunting → narrowest, Local+Additive → broadest.

**Shunting is closest to the biological references considered here.** BP+Shunting produces  $\sigma = 0.94$  on MNIST, very close to the Song et al. reference ( $\Delta = 0.008$ ). Local+Shunting produces  $\sigma = 1.13$ – $1.22$ , close to MICrONS E→E synapse sizes ( $\sigma = 1.14$ ;  $\Delta \leq 0.08$ ). By contrast, Local+Additive produces  $\sigma = 1.43$ – $1.69$ , broader than any biological reference considered. This pattern is consistent across datasets (MNIST and Fashion-MNIST), suggesting it reflects structural properties of the learning rule rather than task-specific features.

**The match depends on inhibitory conductance.** We test sensitivity to the E/I synapse ratio using a sweep over  $N_I \in \{0, 2, 5, 10, 20, 30, 40\}$  inhibitory synapses per branch (Appendix Fig. S17). Without inhibition ( $N_I=0$ ), all conditions produce  $\sigma > 2.0$ , far from biology. With biologically constrained ratios ( $N_I=10$ – $20$ , corresponding to E/I ratios of 4:1 to 2:1; [42]), shunting networks enter the biological range ( $\sigma \approx 0.9$ – $1.2$ ), while the effect of  $N_I$  on additive networks is less systematic. The weight distribution shape is also robust to dendritic depth (branch factors [9] through [3,3,3,3]; see Appendix Fig. S17A): BP+Shunting converges toward Song et al. with increasing depth, while Local+Shunting remains near MICrONS E→E across depths.

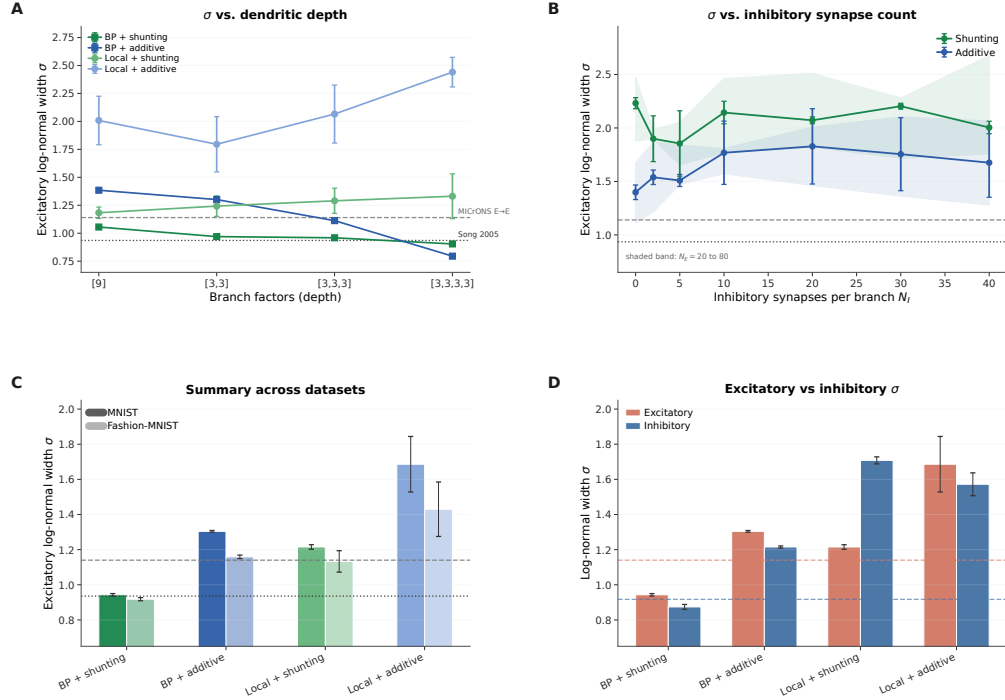


Figure S17: **Synaptic weight distribution sensitivity to architecture.** (A) Log-normal  $\sigma$  vs. dendritic depth (branch factors [9] to [3,3,3,3]). Both backprop and local shunting remain closer to the biological-width references than their additive counterparts across depth. (B)  $\sigma$  vs. inhibitory synapses per branch ( $N_I$ ). Without inhibition ( $N_I=0$ ), all conditions are overly broad; with biologically constrained  $N_I=10$ –20, shunting enters the biological range. Shaded bands show the sweep over  $N_E=20$  to 80. (C) Summary of MNIST and Fashion-MNIST results with biological references. The shunting ordering is consistent across datasets. (D) Excitatory vs. inhibitory weight  $\sigma$  on MNIST with MICrONS E→E and I→E reference bands.

## 728 H Weight Distribution Sensitivity Analysis

729 **Side-by-side BP vs. LocalCA across depths.** To complement the per-architecture analysis above,  
 730 we ran a matched BP vs. LocalCA sweep over three branch-factor settings ([2, 2], [3, 3], [3, 3, 3])  
 731 under both shunting and additive cores on MNIST (5 seeds per cell; quantile init; learned ( $b, m$ ) for  
 732 LocalCA). Figure S18 shows that at shallow and medium depths the local rule and backprop reach  
 733 broadly similar excitatory weight means ( $\approx 0.8$ –0.9), while the coefficient of variation (CV) under  
 734 LocalCA is roughly twice as large as under BP at these depths (e.g. shunting [2, 2]:  $CV_{BP} = 0.15$   
 735 vs.  $CV_{Loc.} = 0.31$ ; shunting [3, 3]: 0.11 vs. 0.26). At the deepest setting [3, 3, 3] the weight means  
 736 drop and the CV jumps substantially for both optimizers ( $CV \approx 0.66$ –0.88), and test accuracy drops  
 737 by  $\sim 5$  pp for BP and  $\sim 15$ –20 pp for LocalCA. Across the explored depth range, shunting LocalCA  
 738 stays closer to shunting BP than additive LocalCA does to additive BP on both weight and accuracy  
 739 axes, consistent with the manuscript claim that the conductance denominator is the key mechanistic  
 740 ingredient.

741 **Weight distributions under soma-on.** To check whether the direct somatic synapse extension  
 742 (Fig. S13) alters the weight-distribution picture, we ran a matched 5-seed sweep at depth [3, 3] on  
 743 MNIST with `somatic_synapses=true` across all four (core, strategy) cells; results are summarized  
 744 in Fig. S19. Two effects emerge. First, shunting + BP becomes *tighter* than in the soma-off case:  
 745 excitatory-weight CV drops from  $\approx 0.11$  (soma-off, [3, 3]) to  $\approx 0.10$  (soma-on), consistent with the  
 746 soma sink absorbing part of the drive and reducing per-branch variance. Second, shunting + LocalCA  
 747 goes in the opposite direction: CV grows from  $\approx 0.26$  (soma-off) to  $\approx 0.40$  (soma-on), while MNIST  
 748 test accuracy drops  $\sim 3$  pp (from 96.5% to 93.6%). Additive cores sit between these extremes (CV  
 749  $\approx 0.23$ –0.30, accuracy 95.7–97.0%) and are only weakly affected. The ordering is consistent with

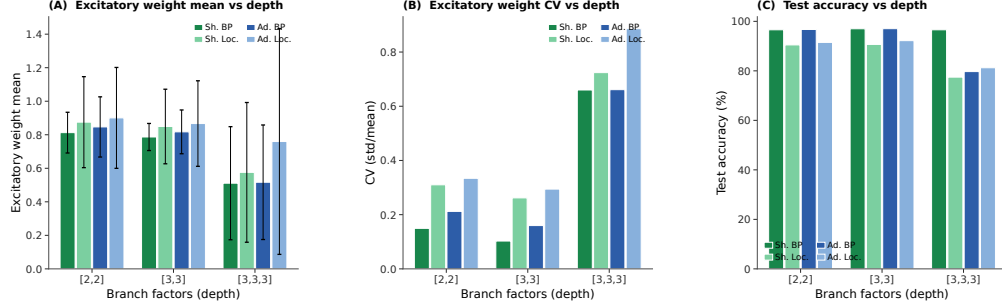


Figure S18: **Synaptic weight statistics vs. dendritic depth (MNIST, 5 seeds/cell).** (A) Excitatory weight mean (error bars = std). At [2, 2] and [3, 3], all four conditions cluster around 0.8–0.9; at [3, 3, 3] the means drop to  $\approx 0.5$ –0.75 and the cross-seed spread grows, especially for additive LocalCA. (B) Coefficient of variation (CV = std/mean). At shallow/medium depth, LocalCA produces  $\sim 2\times$  higher CV than matched BP, but both remain well-behaved ( $\leq 0.33$ ). At the deepest network all four conditions jump to CV  $\approx 0.66$ –0.88, with additive LocalCA the broadest. (C) MNIST test accuracy. BP tracks  $\sim 96$ –97% at [2, 2]/[3, 3] and  $\sim 92$ –95% at [3, 3, 3]. LocalCA stays within  $\sim 5$ –6 pp of BP at shallow/medium depth but trails by  $\sim 15$ –20 pp at [3, 3, 3], identifying depth as the remaining open challenge for this local rule. The qualitative ordering (shunting LocalCA tracks shunting BP more closely than additive LocalCA tracks additive BP) reinforces the manuscript claim that the conductance denominator is the key mechanistic ingredient.

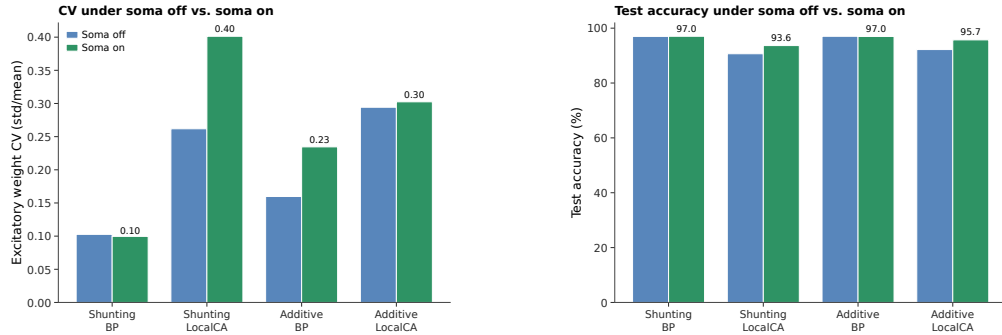


Figure S19: **Excitatory-weight CV and MNIST accuracy under soma-off vs. soma-on (depth [3, 3], 5 seeds/cell).** (A) Coefficient of variation of excitatory weights for the four (core, strategy) cells; paired bars show soma-off (from Fig. S18) and soma-on (this sweep). Shunting + BP tightens slightly; shunting + LocalCA broadens; additive cells change little. (B) Matched MNIST test accuracy (sanity panel). BP accuracy is preserved or marginally improved; LocalCA accuracy is preserved for additive but drops  $\sim 3$  pp for shunting, consistent with the CV widening in panel A.

750 the mechanism in the main text: the somatic synapse is an additional BP-visible path that backprop  
751 exploits cleanly, but under the local rule it adds broadcast variance that the scalar/per-soma signal  
752 cannot sharpen, widening the weight distribution without a matching accuracy gain.

753 **MICrONS connectome analysis.** We extracted synapse size distributions from the MICrONS  
754 mm<sup>3</sup> condensed connectome (v1181; 9.0M connections, 71.9K neurons; 89% excitatory). Split-  
755 ting by pre/post cell type: E→E connections ( $n=3.85$ M) show  $\sigma_{\text{size}}=1.14$  and CV=1.01; I→E  
756 ( $n=3.53$ M) show  $\sigma=0.92$  and CV=1.03; E→I ( $n=1.11$ M) show  $\sigma=0.83$ ; I→I ( $n=0.51$ M) show  
757  $\sigma=0.85$ . Inhibitory connections are more frequently multi-synaptic (I→E: 37.5% multi-synapse;  
758 E→E: 11.5%), and inhibitory synapses are on average smaller (median 3532 vs. 4676 for E→E).

759 **Sensitivity to ee\_synapses.** Using the E/I grid sweep ( $N_E \in \{20, 40, 80\}$  and  $N_I \in$   
760  $\{0, 2, 5, 10, 20, 30, 40\}$ ; 168 models), excitatory  $\sigma$  increases modestly with  $N_E$  in both architec-  
761 tures, but the shunting-versus-additive ordering is preserved at every  $N_E$  value. The dominant driver  
762 of  $\sigma$  is still  $N_I$ : moving from  $N_I=0$  to  $N_I=10$  reduces shunting  $\sigma$  by roughly 1.0–1.5 units, while  
763 the effect on additive is smaller and less systematic.

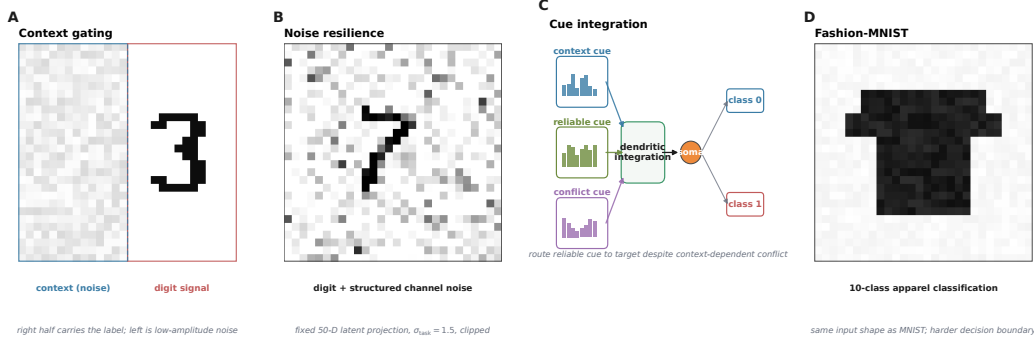


Figure S20: **Synthetic benchmark schematics.** (A) Context gating: the right half carries the MNIST digit; the left half is filled with low-amplitude uniform noise. (B) Noise resilience: MNIST image corrupted by structured channel noise (fixed 50-D latent projection,  $\sigma_{\text{task}} = 1.5$ , clipped to  $[0, 1]$ ). (C) Cue integration: a context cue selects which of two noisy cue streams is reliable; dendritic integration routes the reliable cue to the binary readout. (D) Fashion-MNIST: 10-class apparel classification sharing the MNIST input shape.

**Sensitivity to depth.** From the depth-scaling sweep ( $[9]$ ,  $[3,3]$ ,  $[3,3,3]$ ,  $[3,3,3,3]$ ; 80 models, 5 seeds each), BP+Shunting excitatory  $\sigma$  decreases with depth ( $1.06 \rightarrow 0.90$ ), converging toward Song et al. (0.94). Local+Shunting  $\sigma$  increases slightly ( $1.18 \rightarrow 1.33$ ) but remains in the biological range at shallow depths. BP+Additive shows the strongest depth sensitivity ( $1.38 \rightarrow 0.79$ ), while Local+Additive diverges with depth ( $2.01 \rightarrow 2.44$ ). The morphology-aware modulators (path propagation, depth/centrality weighting) have negligible effect on  $\sigma$ .

**Mechanistic interpretation.** The shunting conductance  $g_n^{\text{tot}} = g_n^{\text{leak}} + \sum g_n^{\text{exc}} + \sum g_n^{\text{inh}}$  appears in the update factor  $R_n^{\text{tot}} = 1/g_n^{\text{tot}}$ . As inhibitory conductance increases,  $g_n^{\text{tot}}$  grows and  $R_n^{\text{tot}}$  shrinks, compressing the multiplicative scale of weight updates. This is analogous to a learning rate that self-adjusts via the total conductance: strong synapses contribute more to  $g_n^{\text{tot}}$ , reducing their own future update magnitude. This acts as a natural form of weight-dependent regularization. The resulting equilibrium distribution is log-normal, consistent with Loewenstein et al. [41] and related multiplicative spine-dynamics analyses. Additive integration lacks this self-normalizing mechanism, producing broader distributions that fall outside the biological range.

## I Synthetic Benchmark Descriptions

Each synthetic task is generated procedurally from a fixed random seed and uses fixed train/val/test splits for reproducibility. The MNIST-derived synthetic tasks use flattened  $28 \times 28$  inputs; cue integration instead uses a small vector input with two output classes. Output dimensionality therefore varies by task and is stated in the task description when it differs from 10-way digit classification. Figure S20 gives a visual summary of each benchmark.

**Context gating.** Inputs are contextual figure-ground MNIST examples. The right half of the image carries the original MNIST digit, while the left half is replaced by low-amplitude uniform noise; all pixels therefore remain nonnegative. This task tests whether credit assignment can exploit a simple context-dependent spatial structure without leaving the nonnegative-drive regime of the conductance model. The HSIC auxiliary objective (weight 0.01) is used to decorrelate layer representations, adding roughly 3 pp under local learning.

**Noise resilience.** Inputs are flattened MNIST images corrupted by structured channel noise: a fixed 50-dimensional latent Gaussian perturbation is projected into input space and added to each example, with  $\sigma_{\text{task}} = 1.5$ , then clipped to  $[0, 1]$ . The projection is fixed across training, so the network must filter correlated interference to recover the underlying class signal. Credit assignment must remain coherent under the corrupted signal, making this task particularly sensitive to path-gain dispersion.

**Info shunting.** Inputs contain two additive streams: a low-frequency signal and a high-frequency distractor drawn with equal energy, together with a separate context channel indicating which stream is relevant on each trial. This benchmark uses signed synthetic inputs, so it violates the nonnegative presynaptic-drive assumption enforced in the main experiments. We therefore retain it only as an exploratory stress test and exclude it from the main-text claims.

**Cue integration.** Two noisy cue streams ( $A$  and  $B$ ) are presented simultaneously, each carrying partial information about one of 2 classes. A binary one-hot context signal indicates which cue is reliable on each trial (cue  $A$  or cue  $B$ ), and each cue vector is clipped to  $[0, 1]$  after noise is added. The fixed-pathway variant duplicates the context into both cue branches; the learned-routing variant requires the router to discover the cue separation from data. This task is used in Sec. A to test structured pathway-vector broadcast while staying inside the nonnegative-input regime.

## J Depth Scaling and Noise Robustness

**Depth scaling.** Varying dendritic depth from 1–4 layers (branch factors  $[9]$  to  $[3, 3, 3, 3]$ ): shunting local degrades from 63.5% to 57.4%; additive local degrades more steeply from 54.9% to 29.7%. The shunting advantage grows with depth ( $+8.5 \rightarrow +27.7$  pp). Backprop ceilings remain stable at about 90–92%. See Fig. S5A.

**Noise robustness.** Gaussian noise  $\mathcal{N}(0, \sigma^2)$  on broadcast error: shunting is robust to  $\sigma \leq 0.1$  (about 62%) while additive drops from 46.5% to chance at  $\sigma=1.0$ , confirming that shunting credit signals carry useful learning information beyond the broadcast magnitude. See Fig. S5B.

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